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The Path to 2075 — Is Convergence Really Over?

Kevin Daly
+44(20)7774-5908 | kevin.daly@gs.com
Goldman Sachs International

- Do poor economies grow faster than rich economies? Many economists argue that the evidence for convergence between low- and high-income economies is weak, suggesting that generalized convergence was largely confined to the 'hyper-globalization' decade of 2000-10. If true, this would be bad news for global well-being, and it would fundamentally weaken a key justification for long-term investment in emerging markets.
- We argue that previous studies that reported only limited evidence of convergence are incomplete, because they fail to weight countries by population size – effectively treating China and Cyprus's experiences as equally significant. With this simple adjustment, we find that economic convergence has been a consistent feature of the global economy since the 1980s.
- The key drivers of this convergence have been Asia's most populous nations: China (since the early 1980s), and India, Indonesia and Bangladesh (from the late 1990s onwards). Together, these four countries account for more than 40% of the global population.
- Among developed economies, the US has been an important positive outlier. Despite possessing one of the world's highest levels of GDP per capita, its GDP per capita growth has continued to outpace most other developed markets.
- The contrast that we find with unweighted convergence studies suggests an intuitive point: periods of rapid globalization, such as 2000-10, are especially beneficial for the development of small economies with limited domestic markets. However, for large emerging market economies – especially those in East Asia – economic convergence appears relatively robust. This ongoing convergence implies that the share of global GDP represented by emerging markets (and East Asian EMs in particular) will continue to rise over time.
- Looking ahead, risks exist in both directions. The development of AI seems likely to bolster the strong performance of the US and parts of East Asia (in particular, China, South Korea and Taiwan). On the downside, increased protectionism could result in a reversal of globalization, while climate change poses a particular risk to Southern Asia and parts of Africa.

Economic Convergence, Led by Asia's Powerhouses

In this *Global Economics Analyst*, we review the performance of GDP per capita across 121 developed (DM) and emerging market (EM) economies since 1980, with a particular emphasis on the process of long-term economic convergence.

The principle of convergence – that poorer economies generally grow faster than richer economies over time – is a central assumption underpinning our *Path to 2075* projections for the global economy (see Box 1).[1] The logic is that the marginal return on investment should be higher in poorer economies where capital is relatively scarce, which attracts investment, increases capital stock, and raises living standards.

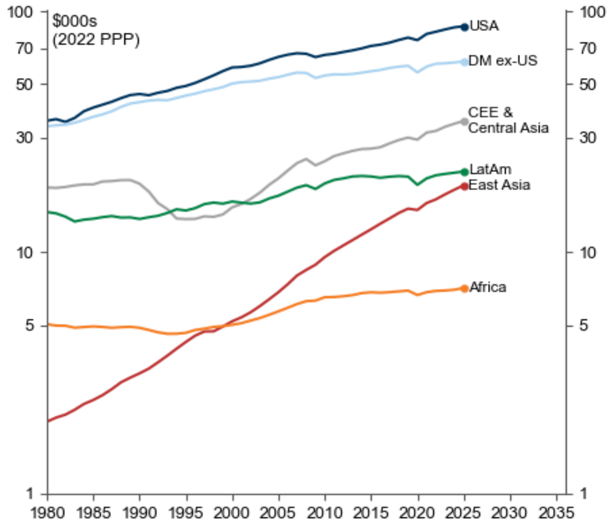
We begin by outlining some key stylized facts about GDP per capita levels and growth rates across the world. Exhibit 1 (left-hand side) depicts GDP per capita across various regions since 1980, while the right-hand panel presents the same data indexed to 1980. Several features stand out:

1. Cross-country differences in GDP per capita levels are large, even after adjusting for differences in purchasing power parity.[2] To take one striking example, GDP per person in the US is more than 12 times higher than the average across Africa.
2. East Asia's emerging markets have seen rapid GDP per capita growth, with average output per person now nearly 10 times higher than in 1980. However, GDP per capita in the region remains well below US and Western European levels, suggesting further potential for convergence.
3. US GDP per capita growth since 1980 has either matched or exceeded GDP per capita growth in the majority of other regions (with East Asia the only exception), despite having a much higher starting level of GDP per capita.
4. GDP per capita growth in Latin American and African economies has disappointed, averaging +0.9% and +0.8% per year respectively since 1980.
5. The break-up of the Soviet Union and the dissolution of the Eastern Bloc was initially very negative for GDP per capita in Eastern Europe and Central Asia in the early 1990s. However, GDP per capita levels have since rebounded in this region.

1 "The Path to 2075 — Slower Global Growth, But Convergence Remains Intact", Goldman Sachs Global Economics Paper, 6 December 2022.

2 The choice between PPP weights and market weights depends on the comparison's purpose. PPP weights account for lower prices in poorer economies and better reflect welfare gains, while market weights show the Dollar value of each economy's GDP without adjusting for spending power. Since we are focused on income convergence, we use PPP weights. See "How Fast Does the World Grow?", *GS Global Economics Analyst*, 27 October 2021.

Exhibit 1: GDP per Capita in East Asian EMs Has Risen Nearly Tenfold Since 1980, But Remains Well Below DM Levels
LHS: Real GDP Per Capita ($000s log scale; 2022 PPP); RHS: Real GDP Per Capita (2022 PPP, 1980=100)



Regions are weighted by population size

Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

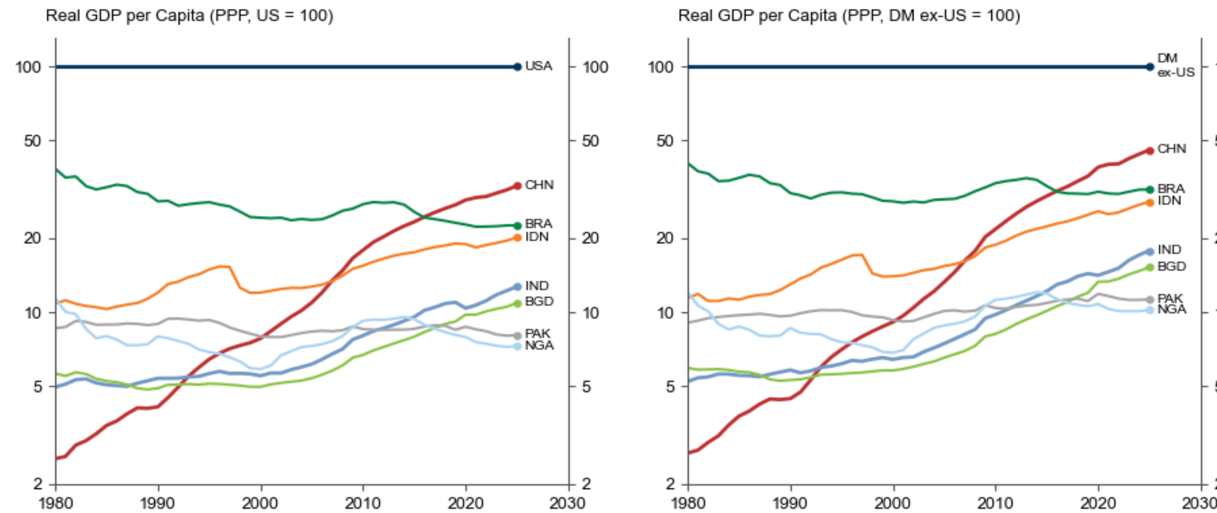
Exhibit 2 compares the GDP per capita of the eight most populous countries. The US is fixed as the 'frontier economy' in the left panel, while developed economies (excluding the US) are set as the benchmark in the right panel. China has seen significant income convergence, rising from just over 2% of US levels in 1980 to around 33% today. Since the late 1990s, India, Indonesia, and Bangladesh have also shown clear progress.

When compared to US GDP per capita, Brazil, Pakistan, and Nigeria have underperformed, with Brazil slipping further behind the US over time, while Pakistan and Nigeria have remained largely unchanged. However, as the US has shown particularly strong performance among developed markets, the relative picture shifts if we compare these economies to developed markets more broadly – Pakistan and Nigeria now show some convergence, whereas Brazil has remained more or less flat.[3]

[3] The aim of this piece is to record, rather than explain, the convergence that has occurred. However, the question remains as to why the GDP per capita performance of Brazil, Pakistan and Nigeria has been disappointing. In Brazil's case, macroeconomic instability, high government spending crowding out private investment, significant trade barriers and inadequate infrastructure are all cited as contributing factors. For Pakistan and Nigeria, macroeconomic instability, elite capture and incomplete property rights are frequently mentioned.

Exhibit 2: China Has Converged Fastest Among the World's Most Populous States, But India, Indonesia and Bangladesh Are Also Converging

Real GDP Per Capita in 2022 PPP, log scale (LHS: US = 100, RHS: DM ex-US=100)



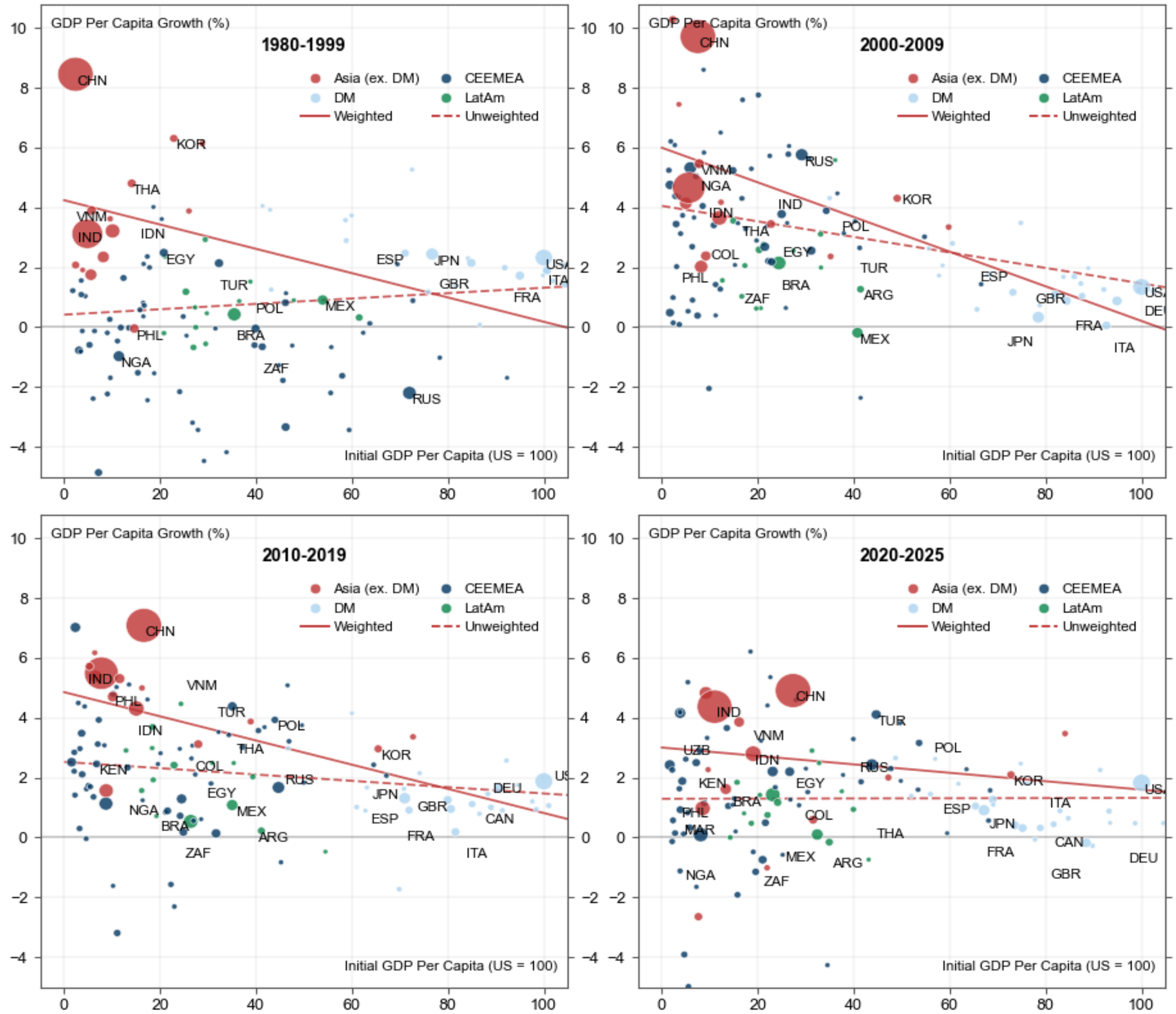
Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

Population-Weighted Convergence is Reasonably Stable Across Decades

Exhibit 3 illustrates average GDP per capita growth in relation to initial GDP per capita levels for EM and DM economies across four distinct periods: 1980-1999, 2000-2009, 2010-2019, and 2020-2025. The level of GDP per capita is shown relative to the US (US = 100). We also show lines of best fit for all 121 economies on both a population-weighted and an unweighted basis.

Across all four periods, the population-weighted regression line slopes downwards, indicating that GDP per capita growth has been higher in low-income economies than in high-income economies (i.e., that there has been convergence). In contrast, the regression lines for unweighted countries are mixed, showing convergence in some periods but not in others. Moreover, the slope of the regression line is steeper for population-weighted countries in all four periods. Note that, although the performance of China and India is central to the narrative of population-weighted convergence – unsurprising given that together they comprise more than a third of the world's population – the regression lines continue to slope downwards from 2000 even when these two countries are excluded.

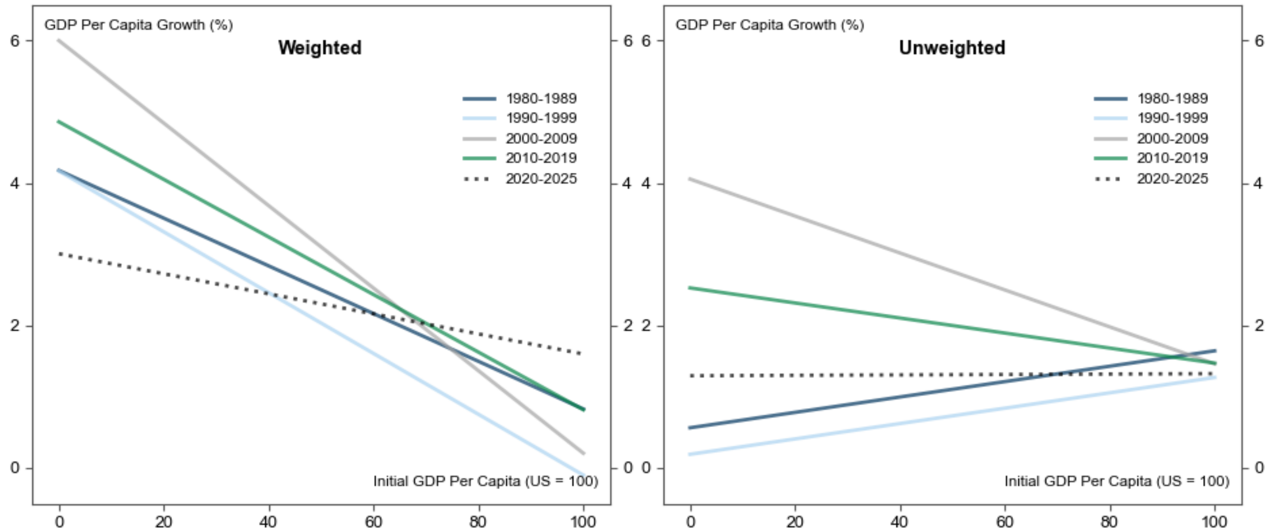
Exhibit 3: GDP per Capita Has Grown Faster in Low-Income Economies than in High-Income Economies
Real GDP Per Capita (PPP, %) vs Initial GDP Per Capita (US=100); Size of bubble = average population size



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

Exhibit 4 examines decade-by-decade changes in regression slopes. The population-weighted regressions show broadly similar slopes, except for 2020-2025, where the slope is still downward but less marked. The past half decade has seen a series of major shocks – the pandemic, the 2022 energy crisis, and 2025 US tariffs – and it remains to be seen how the slope will look for the decade as a whole. In comparison with the weighted lines, the regression slopes for countries on an unweighted basis display much greater variability.

Exhibit 4: GDP per Capita Growth Has Been Faster in Low-Income Economies in Every Decade on a Weighted Basis, But the Pattern is More Mixed on an Unweighted Basis
Population weighted vs unweighted fitted lines



Source: The Conference Board Total Economy Database™ (April 2026), Goldman Sachs Global Investment Research

How rapid is this convergence? Since 2000, individuals in countries with a GDP per capita at 50% of the US level have, on average, experienced GDP per capita growth approximately 2.3pp higher than those in developed economies with productivity levels similar to the US.[4] Over a 10-year period, GDP per capita levels for such individuals would have risen from 50% of US levels to around 61% of US levels.

How Our Results Compare with the Existing Growth Literature

On one level, our finding that the process of economic convergence is relatively robust and has been a consistent feature since the early 1980s should not be particularly surprising, as it is a central prediction of many economic models.[5] However, the majority of cross-country studies have found little evidence that poor countries grow faster than rich ones, irrespective of other factors – in other words, there is little evidence of absolute or generalized convergence in the GDP per capita levels of countries over time.[6]

By making the simple adjustment of weighting each country by population size, our results suggest that convergence between poor and rich economies is a more robust feature than implied by previous studies of this type. **If the unit of aggregation is taken to be people, rather than countries – which is what population-weighting achieves – then convergence is alive and well.**

4 The slope of the regression line for 2000-2025 is around -0.05.

5 Standard (neo-classical) growth theory suggests that relatively poor, low-productivity economies should grow at a faster rate than relatively rich, high-productivity countries. This result follows directly from the assumption of diminishing marginal returns on capital: in developing economies where capital is limited, the returns on capital should be greater than in richer economies where capital is abundant. In turn, the relatively high returns on capital in less developed economies should attract investment, increasing the capital stock and raising living standards.

6 A key paper in this literature is Barro and Sala-i-Martin (1992), 'Convergence', *Journal of Political Economy*. Other studies found evidence of 'conditional' convergence – meaning that convergence occurs only when specific factors, such as institutional quality, are present. See Mankiw, Romer and Weil (1992) 'A Contribution to the Empirics of Economic Growth', *Quarterly Journal of Economics.*

We are not the first to adjust for population size in this way. Previous studies have also weighted by population size and, consistent with our findings, have found stronger evidence of convergence on this basis.[7] Nonetheless, those studies maintain that, when assessing which economic policies drive growth and convergence, it is still reasonable to treat each country as an equal unit, since such policies are largely determined at the national level. We disagree for two reasons: First, economic welfare is a study of people, not nations, and if more people reside in some nations than others, they should be weighted accordingly. Second, a country's size is itself a potentially significant factor influencing growth and convergence. If larger countries tend to grow faster than smaller ones, this may partially be attributable to their size. Overlooking this aspect risks missing a potentially important determinant of growth.[8]

The contrast that we find with unweighted convergence studies suggests an important and intuitive point: periods of rapid globalization, such as 2000-10, tend to be particularly beneficial for the development of small economies with limited domestic markets. However, for large emerging market economies, economic convergence appears to be more resilient to the pace of globalization.

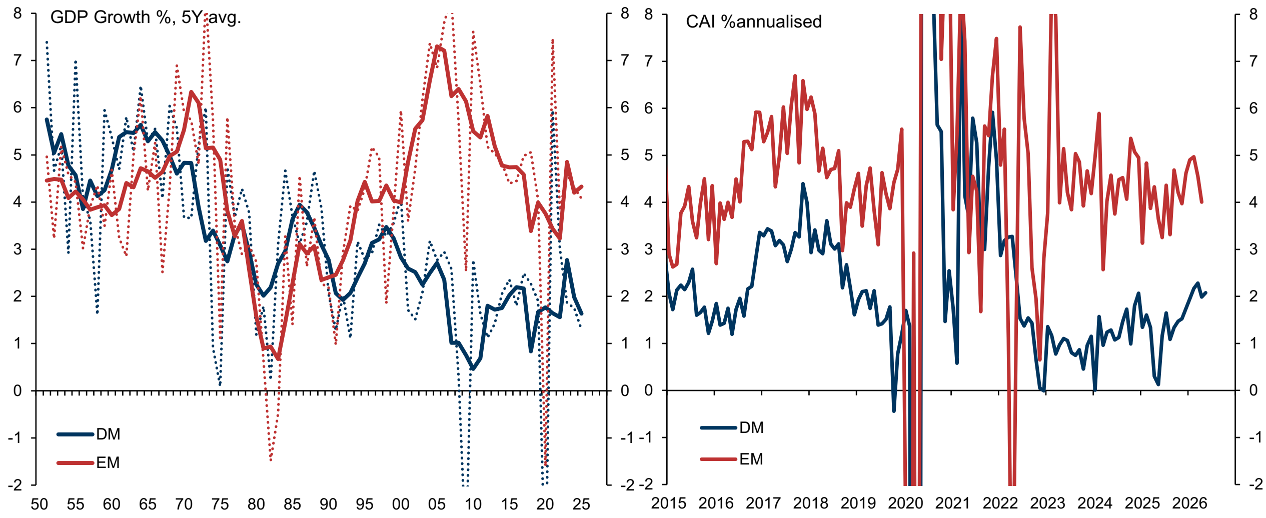
EM Growth Continues to Outstrip DM Growth by a Wide Margin

Our findings also explain why EMs consistently outgrow DM countries in our coverage – if convergence were not a fairly consistent feature of the global economy, this would not occur. While population growth is also higher in EMs, it does not fully explain the gap in potential growth rates (4.2% vs 1.6%). Moreover, in the four years since we published our *Path to 2075* long-term economic projections, the gap between EM and DM growth has so far been larger than we expected (2.6pp vs. 2.0pp).

[7] In a follow-up paper to Barro and Sala-i-Martin (1992)), Sala-i-Martin (2006) acknowledges that there is stronger evidence of convergence in income distribution at the level of the individual rather than the country but maintains that a country framework makes sense when evaluating which economic policies drive growth and convergence. See Sala-i-Martin (2006), 'The World Distribution of Income: Falling Poverty and... Convergence, Period', *Quarterly Journal of Economics*.

[8] A large domestic market makes it easier to achieve economies of scale while still focused primarily on domestic demand, especially if industries are protected during their early stages of development. This strategy – protecting fledgling industries until they reach sufficient size – is less likely to succeed in countries with a small domestic market. There is also a related but distinct point concerning economic geography: large economies are able to support bigger cities and benefit from agglomeration effects, and economic convergence is often associated with rising urbanization and increased specialization. This is more difficult to achieve in small economies with small populations.

Exhibit 5: After a Brief Hiatus During the Pandemic, EM Economies Are Outgrowing DM Economies
LHS: GDP Growth (%, 5Y avg); RHS: Current Activity Indicator (% change, annualized)



Source: Goldman Sachs Global Investment Research

The ongoing convergence between (population-weighted) poor and rich economies implies that the incomes of EM economies will gradually move closer to those of developed economies – resulting in a further reduction in global income inequality – and that the share of global GDP represented by emerging markets will continue to increase over time.

Looking forward, there are both positive and negative risks to long-term economic growth and convergence. The development of AI seems likely to bolster the strong performance of the US and parts of South-East Asia and lift global GDP productivity growth more generally.[9] In terms of AI-related capital expenditure, South Korea and Taiwan are already the main beneficiaries, while from a development and adoption standpoint, China seems well-placed to gain.

On the downside, we consider two risks especially significant: First, although global growth and trade have so far been surprisingly resilient to rising tariffs, increased protectionism could still trigger a reversal of globalization. Second, climate change represents a particular threat to Southern Asia and parts of Africa.

Kevin Daly and **Alessa Abraham***

**Alessa is an intern in the CEEMEA economics team.*

[9] For a discussion of the potential boost to EM productivity from AI and technology adoption more generally, see "EM Technology Adoption Cycles and Implications for Artificial Intelligence", *Global Economics Comment*, 21 May 2026.

Box 1: The Path to 2075 — Long-Term Projections for the Global Economy

The Path to 2075 projections builds on our original BRICs analysis, offering long-term forecasts for the global economy, covering 104 countries out to the year 2075.[10] Exhibit 6 provides a high-level summary of our projections for the major regional aggregates, broken down by decade.

Our projections imply that global growth in 2024-29 (2.8%) will be faster than in 2020-24, but slower than in the 2010-19 decade (3.2%, based on market FX weights).[11] We project that EM growth will continue to outstrip DM growth (3.8% vs. 1.8%), with around half of this difference due to demographic factors and the other half due to productivity growth differentials. We expect Asia (ex-DM) to remain the fastest-growing region.

Exhibit 6: A Gradual Slowdown in Global Economic Growth, With EM Growth Continuing to Outstrip DM

	Market FX Weighted								
	2000-2009	2010-2019	2020-2029	2024-2029	2030-2039	2040-2049	2050-2059	2060-2069	2070-2079
World	2.7	3.2	2.4	*2.8*	2.5	2.1	2.0	1.8	1.7
DM	1.6	1.9	1.5	*1.8*	1.6	1.4	1.3	1.2	1.1
EM	5.7	5.1	3.6	*3.8*	3.2	2.6	2.3	2.1	1.9
Asia (ex. DM)	7.6	6.7	4.1	*4.2*	3.1	2.4	2.1	1.8	1.5
CEEMEA	4.8	3.5	2.6	*3.2*	3.3	3.1	3.0	2.9	2.7
LatAm	2.8	2.4	2.3	*3.0*	3.1	2.7	2.3	1.9	1.6

Source: Goldman Sachs Global Investment Research

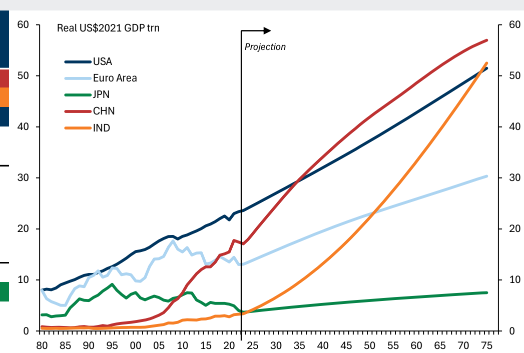
Exhibit 7 combines our GDP estimates with our long-term real exchange rate projections, to project the real US Dollar value of major economies over time.

In 2050, we project that the world's five largest economies will be China, the US, India, Indonesia, and Germany (with Indonesia displacing Brazil and Russia among the list of largest EMs over this horizon). We estimate that China will overtake the US as the world's largest economy around 2035.[12] If we extend the projection horizon to 2075, the world's three largest economies are China, India, and the US, with India (just) overtaking the US. Interestingly, US potential GDP growth is expected to be materially faster than China's at that horizon because of its better demographic outlook. Given the right policies and institutions, we project that current EMs will make up seven of the world's top ten economies in 2075.

Exhibit 7: Our Projections Imply that China, the US, India, Indonesia, and Germany Will be the World's Five Largest Economies in 2050

World's largest economies (measured in USD)

Ranking	1980	2000	2022	2050	2075
1	United States	United States	United States	China	China
2	Japan	Japan	China	United States	India
3	Germany	Germany	Japan	India	United States
4	France	United Kingdom	Germany	Indonesia	Indonesia
5	United Kingdom	France	India	Germany	Nigeria
6	Italy	China	United Kingdom	Japan	Pakistan
7	China	Italy	France	United Kingdom	Egypt
8	Canada	Canada	Canada	Brazil	Brazil
9	Argentina	Mexico	Russia	France	Germany
10	Spain	Brazil	Italy	Russia	United Kingdom
11	Mexico	Spain	Brazil	Mexico	Mexico
12	Netherlands	Korea	Korea	Egypt	Japan
13	India	India	Australia	Saudi Arabia	Russia
14	Saudi Arabia	Netherlands	Mexico	Canada	Philippines
15	Australia	Australia	Spain	Nigeria	France



Source: Goldman Sachs Global Investment Research

10 "The Path to 2075 — Slower Global Growth, But Convergence Remains Intact", Goldman Sachs Global

Making economic projections over a 50-year horizon for 104 countries inevitably involves a considerable degree of risk and uncertainty. Indeed, given the difficulties in forecasting even one or two years ahead, some readers may be skeptical of the value of forecasts that extend so far into the future. However, one advantage of making long-term projections is that cyclical risk – an important source of short-term forecasting error – tends to mean-revert over time, leaving GDP mostly determined by slower-moving trends in population, capital, and technology.

Economics Paper, 6 December 2022.

[11] Because of the large swings in GDP growth generated by the Covid pandemic, the 2024-29 column provides the cleanest indication of what our model indicates for near-term potential growth.

[12] Given the recent pessimism around China's long-term growth prospects, some readers may be surprised that we expect China to overtake the US at this horizon. However, three points are worth bearing in mind in this regard: First, China has already closed most of the gap with US GDP (China's GDP has risen from 12% of the US in 2000 to around 80% currently, measured in market exchange rates). Second, despite significant downward revisions, potential growth in China remains significantly higher than in the US on our revised estimates (4.0% vs. 1.9% for 2024-29). Third, in addition to differences in potential growth, we expect some of the US Dollar's real overvaluation vs. the Chinese Yuan to be unwound over the next 10-15 years.

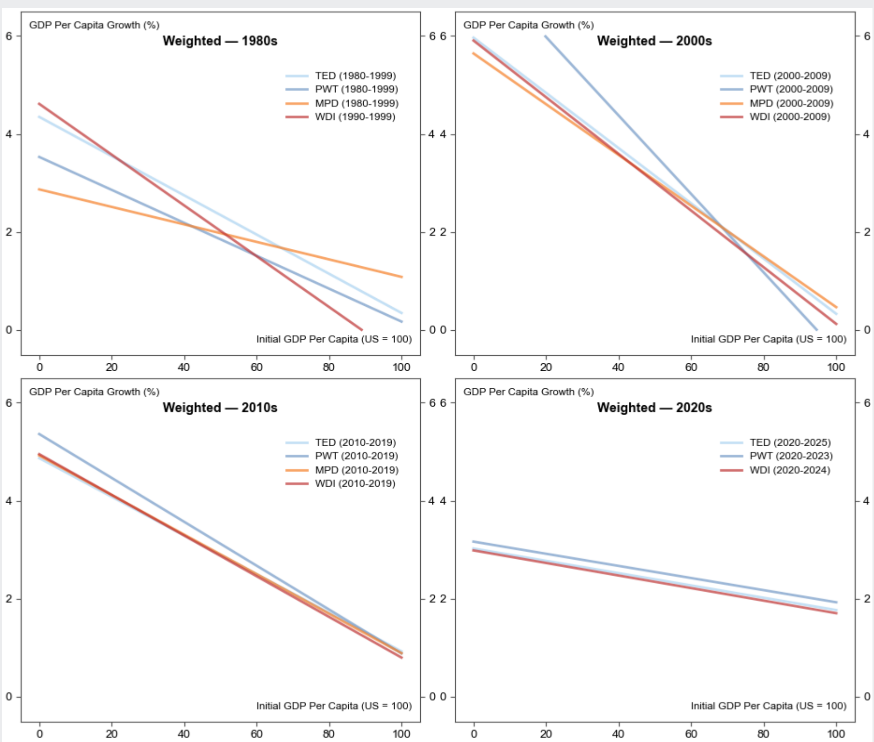
Box 2: Comparing Our Results Across Different Growth Databases

Growth and convergence studies typically use one of four different databases when analyzing long-term economic performance: World Development Indicators (WDI) by the World Bank, the Penn World Tables (PWT), the Maddisson Project Database (MPD), and the Total Economy Database (TED) developed by the Conference Board. We generally use the Total Economy Database, partly because it provides the most up-to-date GDP figures. However, to cross-check our findings, we have replicated our analysis using each of these databases to ensure the robustness of our results.

Each of these databases possesses distinct advantages and limitations, making some more suitable than others depending on the research objectives. Among them, the Maddisson database offers the most extensive historical coverage, in some cases reaching back centuries, making it ideal for studies in economic history. The World Bank's database has the shortest historical span, beginning only in 1990. Another key distinction is how they adjust GDP figures for differences in Purchasing Power Parity (PPP), which in turn impacts growth indicators.

The results of replicating our convergence analysis across each of the databases are presented in Exhibit 8. Although there are minor variations in the slopes of the fitted lines, the principal result—that weighted convergence persists—remains broadly consistent across all datasets.

Exhibit 8: Our Results Are Similar Across Each of the Main Growth Databases
Fitted lines - weighted by population, across the decades



Source: The Conference Board Total Economy Database™ (April 2026), Maddison Project Database (2020), Penn World Table 11.0, World Bank World Development Indicators, Goldman Sachs Global Investment Research

The Global Economics Team

Jan Hatzius
+1(212)902-0394
jan.hatzius@gs.com
Goldman Sachs & Co. LLC

Joseph Briggs
+1(212)902-2163
joseph.briggs@gs.com
Goldman Sachs & Co. LLC

Sarah Dong
+1(212)357-9741
sarah.dong@gs.com
Goldman Sachs & Co. LLC

Megan Peters
+44(20)7051-2058
megan.l.peters@gs.com
Goldman Sachs International

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Contributing Authors: Kevin Daly Goldman Sachs International.

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EUROPEAN ECONOMICS ANALYST

Spain – Structural Resilience Beyond the Energy Shock

Filippo Taddei
+44(20)7774-5458 |
filippo.taddei@gs.com
Goldman Sachs International

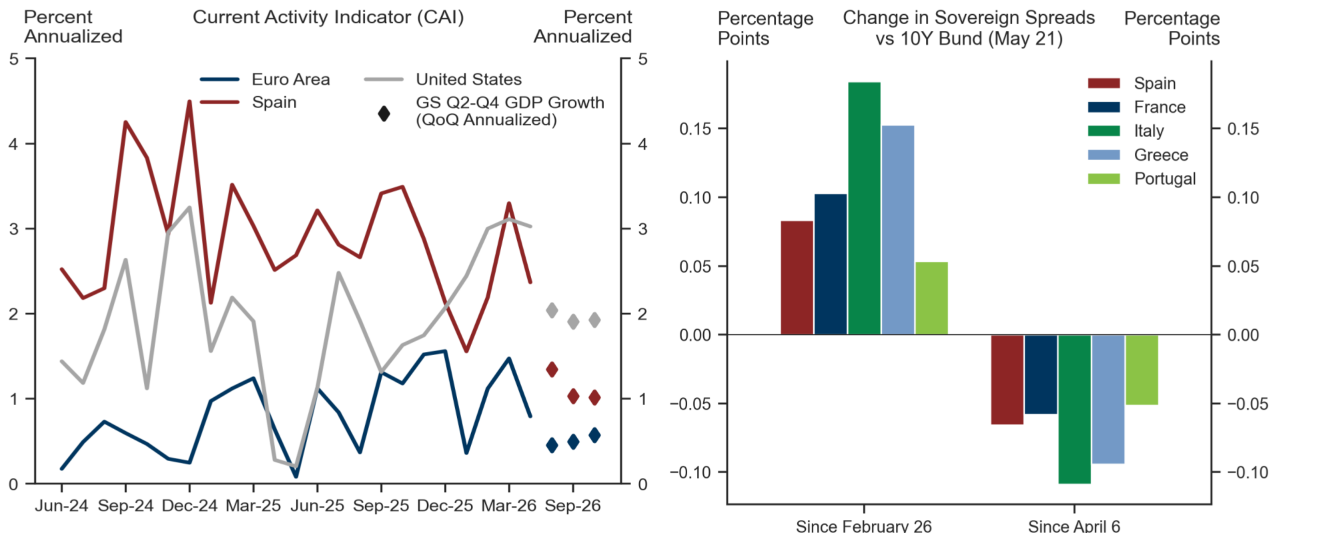
- Although the energy shock is taking its toll on the Spanish economy, we still expect economic growth to outperform the rest of the Euro area. With growth running strongly above trend and the primary fiscal balance at the highest level since 2007, Spanish sovereign spreads are well-supported, relatively tight and resilient.
- The fiscal mix in Spain differs from other EMU4 countries. The government has decided not to prioritise defence spending and, even though the Spanish economy is less exposed to the conflict in the Middle East given its lower energy intensity, its fiscal response to the surge in energy prices is the most aggressive within the EMU4.
- Strong job growth has coincided with strong immigration and a shift in the workforce towards higher value-added service sectors—especially professional services, finance, and information technology—continuing to signal structural progress in the Spanish economy. Since 2021, Spain has recorded the highest productivity growth, per employee and per hour, within the EMU4.
- Investment growth remains resilient and broad-based. Spain's total capital expenditure is back to the level before the sovereign debt crisis and investment as a percent of GDP is set to overtake the Euro area, US and UK in 2026. The composition of the investment recovery also looks promising. The construction sector has consolidated its growth momentum and the surge in residential real estate prices due to net migration should attract additional investment in the sector. The key support from the European Recovery Fund comes to an end this year. However, we estimate that its effect will extend further.
- Against this macro backdrop, investors are constructive on Spanish government bonds. Our analysis finds no pricing of domestic risk: Spanish sovereign spreads have widened only in combination with flatter yield curves across Europe, pointing to rising inflation and slowing global growth as the sole factors responsible for the recent widening.
- We see two main risks that could potentially challenge this constructive outlook. As we move into the Summer, we estimate that an unresolved energy crisis feeding into air travel costs could lower GDP by 0.3% for every 10% reduction in tourist arrivals by air. Looking ahead to the 2027 election, any shift in opinion polls pointing to another split parliament or an increased likelihood of a minority

government would reduce investors' confidence in extended gains on the economic front.

Spain – Structural Resilience Beyond the Energy Shock

Spain's macro outperformance remains visible both in economic activity and in sovereign debt pricing. Economic activity has continued to run ahead of the Euro area and the US since the middle of last year, while sovereign spreads have remained comparatively tight, indicating investors' confidence in the outlook for Spain. That said, the Spanish economy is set to slow down because of the energy shock, although it is still outperforming the Euro area (Exhibit 1).

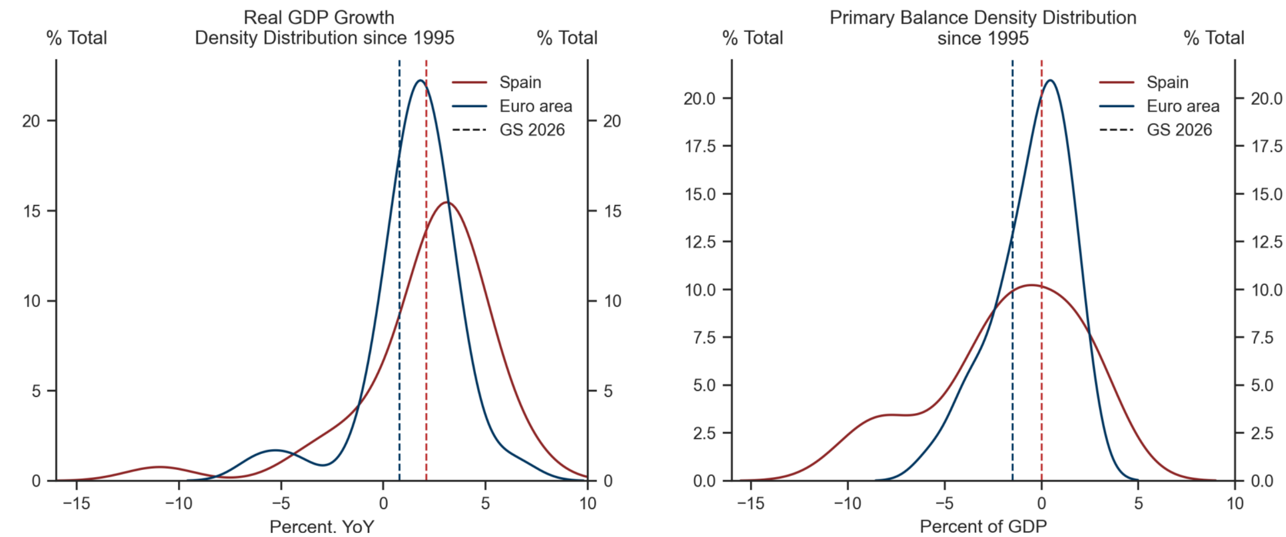
Exhibit 1: Growth Outperformance Keeps Spreads Tight



Source: Goldman Sachs Global Investment Research, Haver Analytics

Current macroeconomic performance compares favourably from an historical standpoint. Spain entered this energy crisis from a relatively robust macro position. Even accounting for the upcoming slowdown, we forecast real GDP growth in 2026 at or above the long-run average, while the primary fiscal balance compares favourably with historical precedents and is the third-highest in the last 19 years (Exhibit 2).

Exhibit 2: Primary Balance and Real Growth are Close or Above the Historical Average

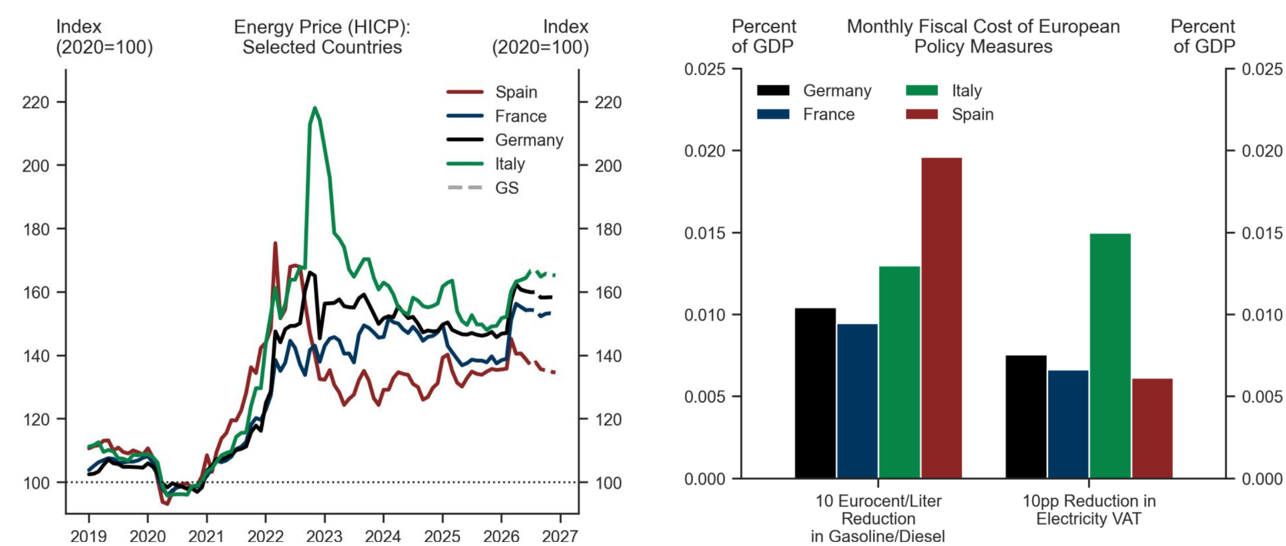


Source: Goldman Sachs Global Investment Research, Haver Analytics

A Relatively Strong Fiscal Response to the Energy Crisis Leaves the Fiscal Balance in a Good Place

Spain's lower energy intensity helps contain the direct hit to growth, while fiscal risks remain manageable despite a relatively significant policy response. Since the 2022 energy crisis, Spain has been the least exposed in the EMU4 to the surge in energy prices. The fiscal cost of the most common measures used to shield consumers also sit at the lower end of the distribution in Spain, with the notable exception of tax cuts for petrol and diesel (Exhibit 3).

Exhibit 3: Lower Energy Exposure, Limited Fiscal Risk

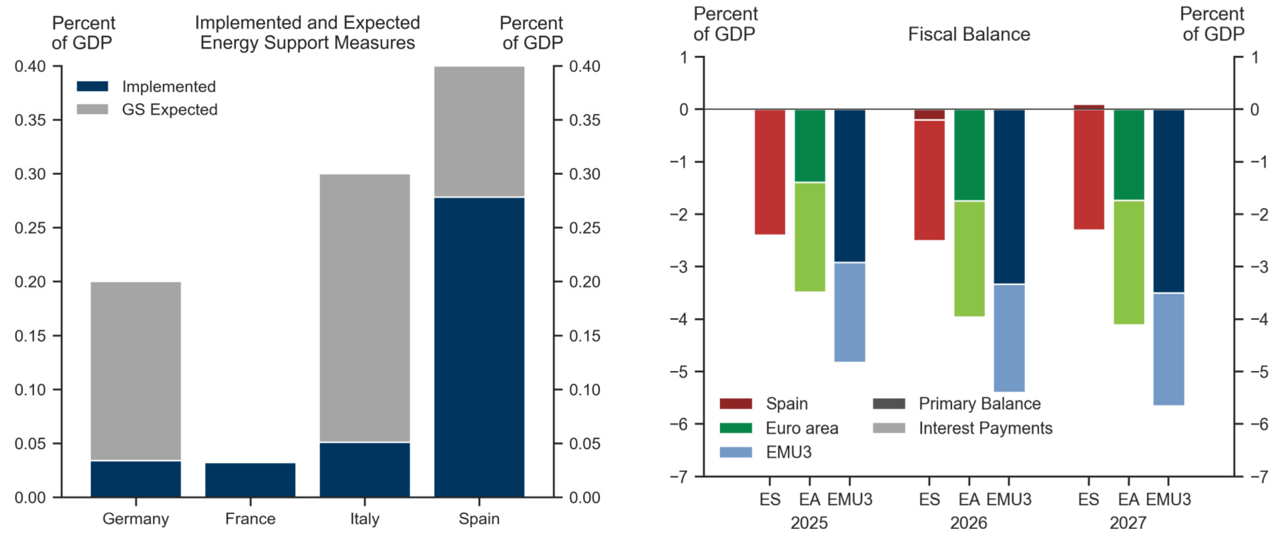


Source: Goldman Sachs Global Investment Research, Haver Analytics

However, Spain has delivered one of the most forceful energy-related fiscal responses in the region. Its decision not to prioritise defence spending and its relative insulation from the shock has allowed policymakers to provide support without materially undermining the broader fiscal position. While the increase in support has been the largest in the

EMU4, the fiscal balance remains among the most constructive in the Euro area. Spain's policy mix has been relatively expansionary, but it has not damaged its credibility from a sovereign perspective as Spain remains the only EMU4 country where we expect the debt-to-GDP ratio to be lower over the next three years (Exhibit 4).

Exhibit 4: Larger Energy Support, Strong Fiscal Position



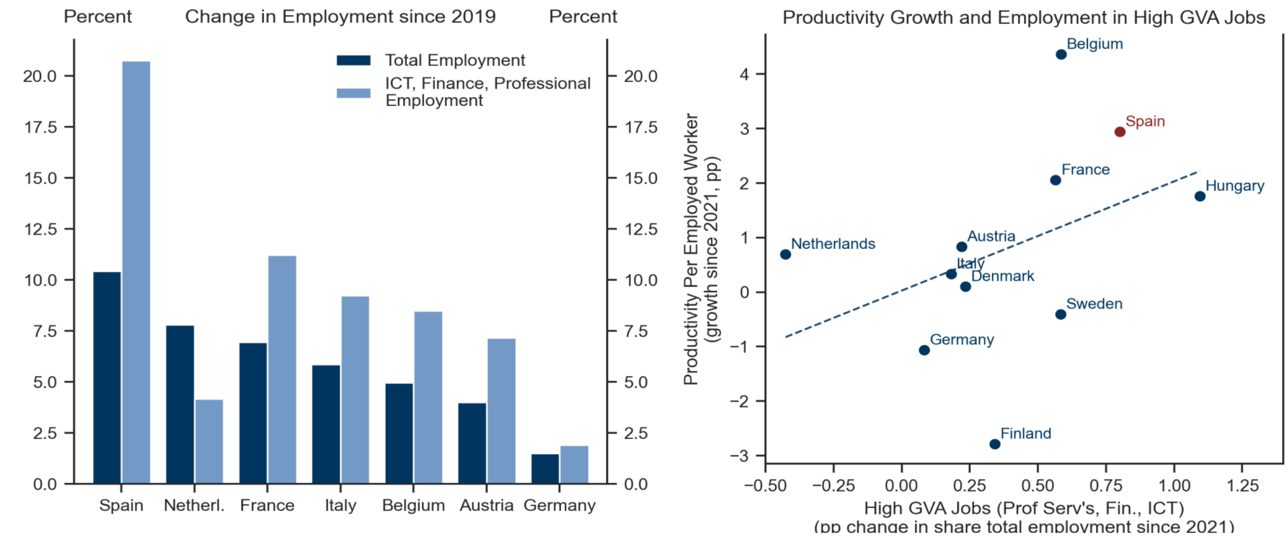
EMU3 includes Germany, France and Italy.

Source: Goldman Sachs Global Investment Research, Haver Analytics, Bruegel

Continuing Evidence of Structural Productivity Improvements

The employment rate is the highest it has ever been and the unemployment rate is the lowest since 2008. The Spanish labour market remains cyclically stable, accommodating large net migration flows, and employment gains have increasingly been concentrated in higher-value-added service sectors (professional services, finance and ICT). This job reallocation, a specific feature of the Spanish economy in the European landscape, appears to be feeding through into stronger productivity performance. Spain's recent outperformance likely reflects a gradual improvement in the quality—not only the quantity—of labour demand (Exhibit 5).

Exhibit 5: Labour Reallocation Supports Productivity

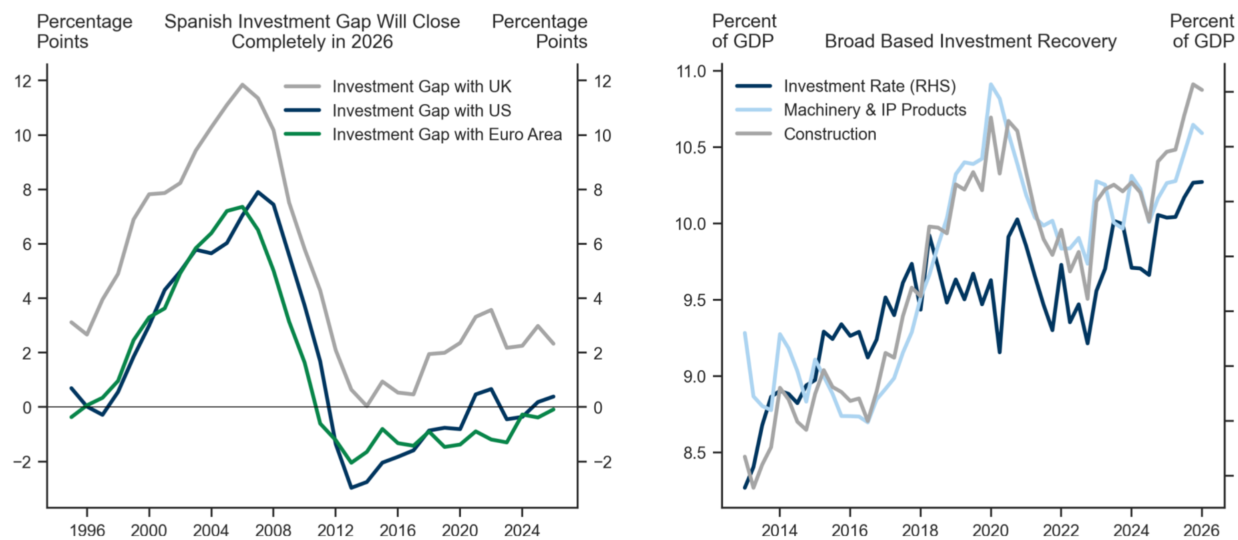


GVA stands for Gross Value Added.

Source: Goldman Sachs Global Investment Research, Haver Analytics

Capital accumulation has also been an engine for growth. The investment recovery in Spain is both resilient and broad. Capital expenditure has regained ground relative to past cycles and is set to close the investment gap with the Euro area, US and UK this year. The composition of the investment recovery looks especially promising, with capital expenditure on machinery and intellectual property products the highest in 25 years. A durable investment momentum is typically associated with an extended expansion (Exhibit 6).

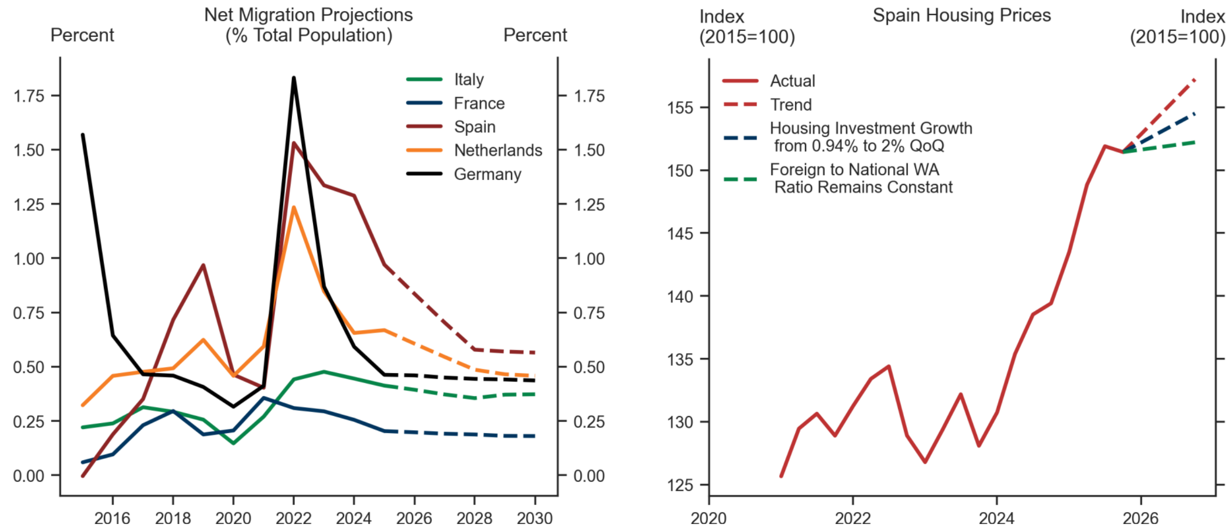
Exhibit 6: Investment Recovery Gains Traction



Source: Goldman Sachs Global Investment Research, Haver Analytics

The construction sector has consolidated its growth momentum. Net migration into Spain in particular has been elevated since 2023 and a key factor supporting economic growth. But demographic pressure has also tightened the housing market, raising sharply residential real estate prices by almost 20% over the last two years. While house prices have risen, supply has struggled to keep pace and we estimate that housing investment needs to double in order to normalize residential real estate appreciation. We expect this factor, along with the surge in residential real estate prices due to net migration, to attract additional investment in the sector (Exhibit 7).

Exhibit 7: Net Migration and Demographics Lift Housing Prices

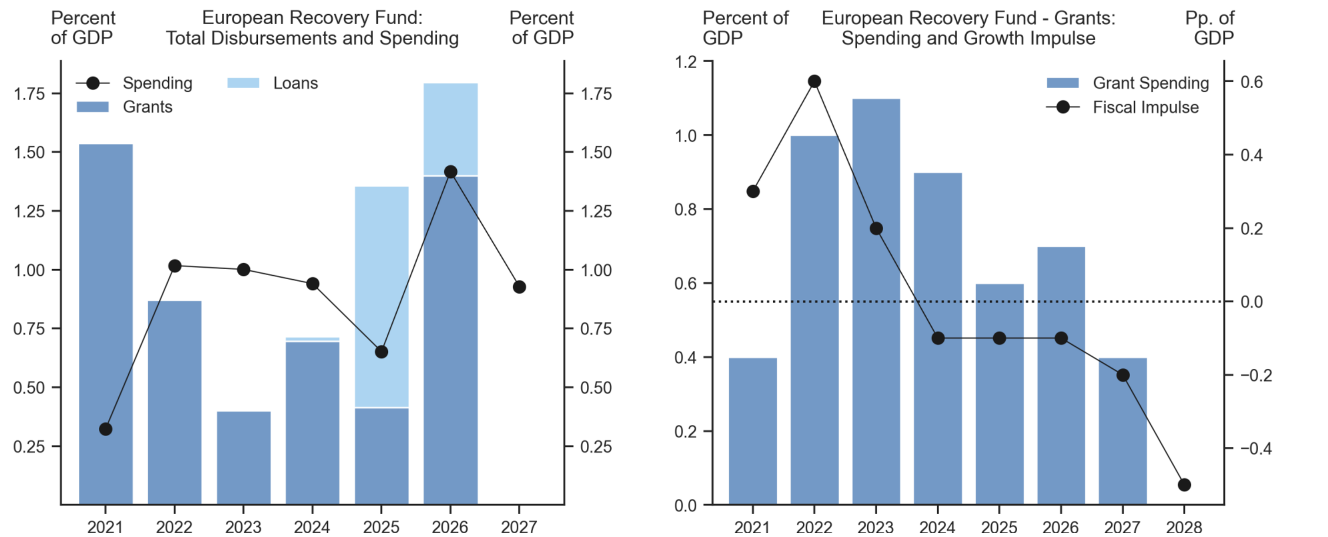


Source: Goldman Sachs Global Investment Research, Haver Analytics

The government has also set up a specific policy tool to support developments in the housing sector. By establishing the *España crece* (Spain Grows) fund, the government plans to mobilise EUR 60bn through the Official Credit Institute (ICO) to reach a total investment of around EUR 120 billion (9% of GDP), combining public and private capital. Endowed with a capital injection of EUR10.5bn from the European Recovery Plan, the fund explicitly targets private leverage to augment investment in residential developments.

The fund was also part of a broader strategy of the Spanish government to extend the impact of the European Recovery Fund beyond the end of the programme in 2026. The Recovery Fund has been a key factor supporting investment growth and comes to an end this year. Since a large share of investment projects will only be completed in the first half of next year and with the *España crece* fund in place, we estimate that the support of the European Recovery Fund will extend beyond the end of 2026, suggesting a moderation rather than collapse of fiscal support for investment, and reducing the risk of an abrupt halt in capex momentum (Exhibit 8).

Exhibit 8: Recovery Fund Support Fades Gradually



Source: Goldman Sachs Global Investment Research, Haver Analytics

A Constructive Outlook for Spanish Debt

Investors are constructive on Spanish bonds as weakening momentum due to the energy crisis seems to have been unaffected by country-specific risks. In line with previous work, we use a simple structural vector autoregression (SVAR) model with "sign restrictions" to disentangle the underlying macro drivers of sovereign debt pricing. To do so, we define the drivers as follows:

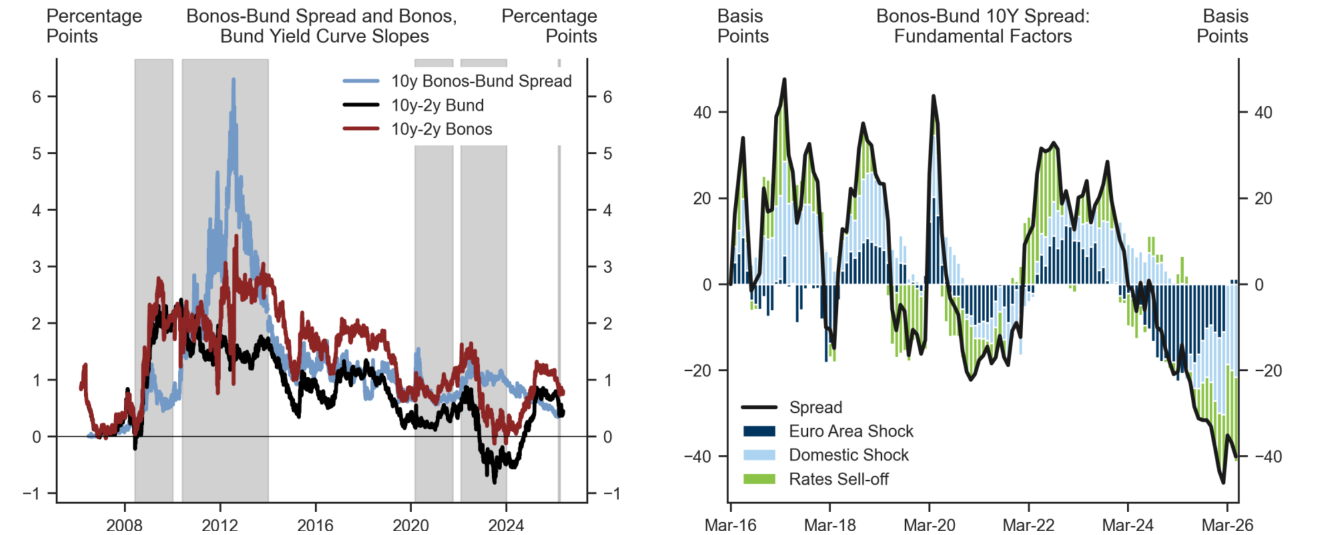
1. Euro Area Shock: when both the Spanish and German sovereign yield curve flatten (10y-2y Bonos and 10y-2y Bund) and the Spanish sovereign spread (10y Bonos-Bund) widens.

2. Domestic Shock: when only the Spanish sovereign yield curve flattens and the Spanish sovereign spread widens at the same time.

3. Long-End Rates Sell-off: when both the Spanish and German sovereign yield curves steepen and the Spanish sovereign spread widens.

In this simple way, it is possible to assess whether the market pricing of Spanish sovereign spreads is more likely to be due to external macro factors – e.g., a deteriorated growth outlook or inflation surging in the region, with rates selling off – or to risk factors specific to the Spanish landscape that push up the yield curve and widen the sovereign spread at the same time. We find that domestic risk (a flatter Spanish curve and wider spread) has remained contained throughout the energy shock so far, while spread widening has been combined with flatter yield curves across Europe. This points to rising inflation and growth concerns beyond the idiosyncratic risks for the Spanish economy and is consistent with the view that investors are currently attaching a negligible Spain-specific risk premium (Exhibit 9).

Exhibit 9: Spreads Price European, Not Domestic, Risk



The sovereign spread depicted in the right chart shows the deviation of the spread form the sample average.

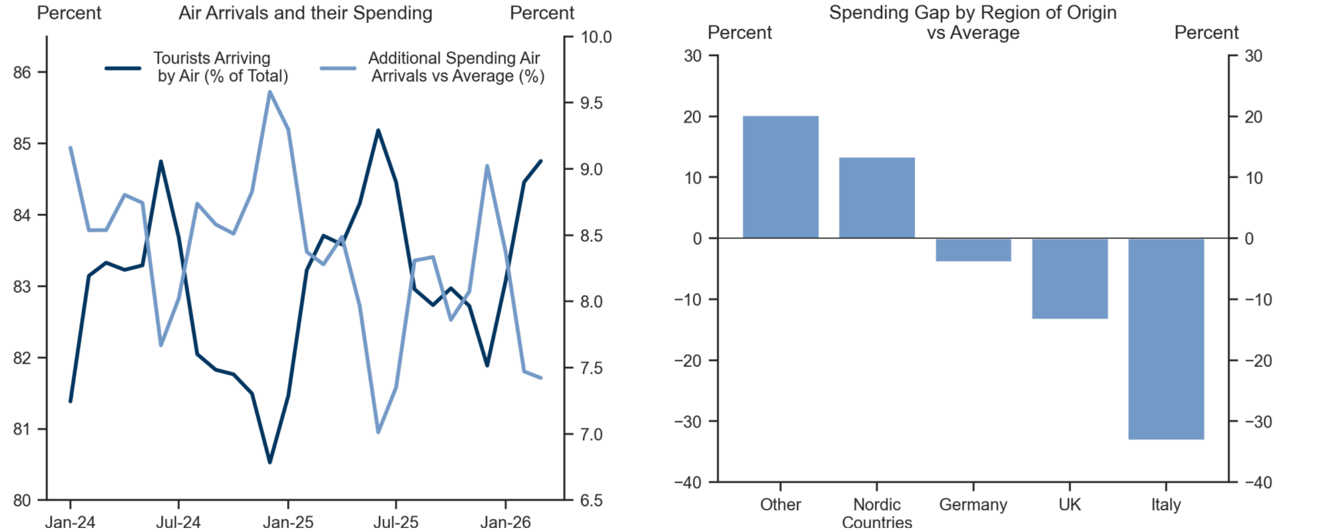
Source: Goldman Sachs Global Investment Research, Haver Analytics

Two Risks To Watch: Energy Hit on the Tourism Sector and Prolonged Political Uncertainty

We see two main risks that could potentially derail the otherwise constructive outlook for the Spanish economy.

The first is the potential impact of the energy crisis on the tourism sector in Spain specifically, and Southern Europe more broadly. As the Summer approaches, an unresolved energy crisis could feed into air travel costs. Tourist arrivals by air matter disproportionately for Spain's growth outlook. They not only represent almost 90% of total international tourist arrivals in Spain but they also display a higher propensity to spend. An energy-driven increase in travel costs would not only reduce arrivals, but could also weigh more heavily on the highest-value segment of external demand (Exhibit 10).

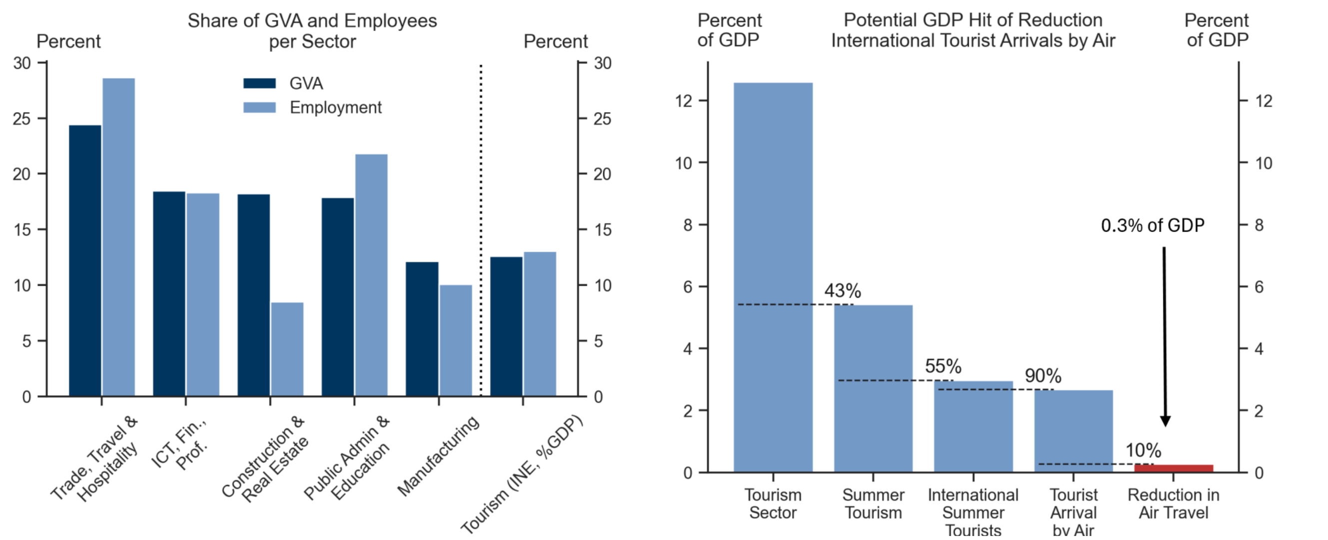
Exhibit 10: Air Travel Brings Higher-Spending Tourists



Source: Goldman Sachs Global Investment Research, Haver Analytics

The size of the tourism sector is not easy to measure because of the multiple spillovers in related sectors, such as trade and cultural services. However, available estimates place the sector in the range of 10-15% of Spanish GDP. The National Statistical Institute (INE) estimates that tourism contributes 12.6% of GDP. With this magnitude in mind, and taking into account the relevance of the summer season (June-September) and the share of international tourist arrivals by air in total spending, we find that every 10% reduction in tourist arrivals by air could lower GDP by about 0.3% (Exhibit 11).

Exhibit 11: Tourism is Large Share Spanish GDP and Tourist Arrivals by Air is Key Component

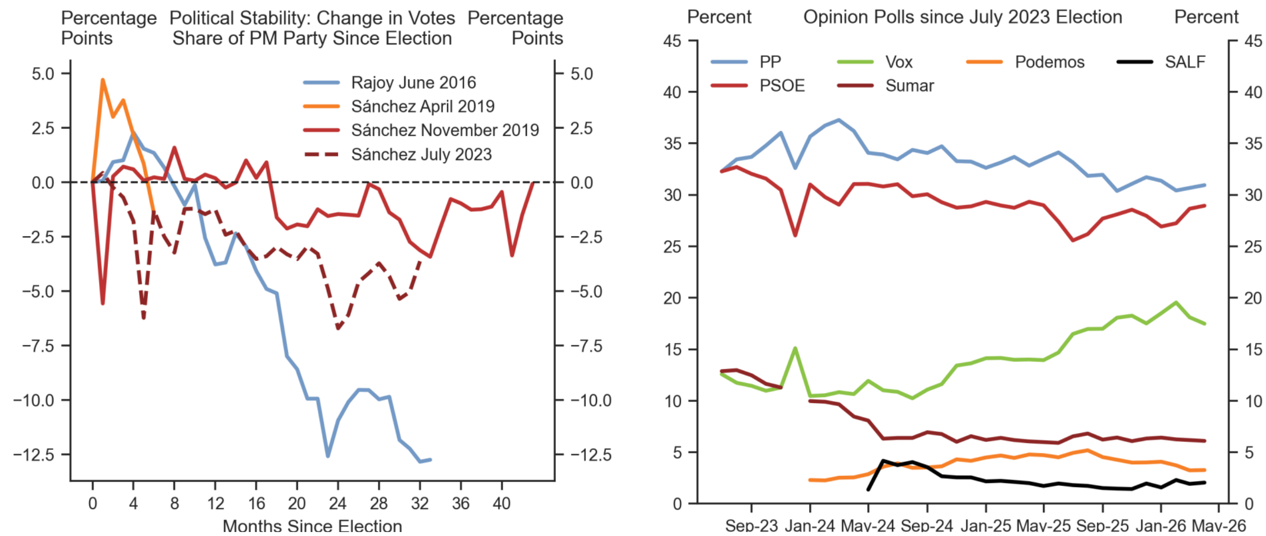


Source: Goldman Sachs Global Investment Research, INE, Haver Analytics

The second risk for the Spanish outlook is prolonged political uncertainty. PM Sanchez has been in office since 2018, but since the 2023 snap election he has led a minority government unable to approve a fully fledged annual budget. This limited ability to shape fiscal policy has not impeded the reduction of Spain's fiscal deficit, but the support of the European Recovery Fund has been key in securing investment growth despite political fragmentation.

With the European programme coming to an end this year, the need for a government able to approve the annual budget will become more urgent. Current opinion polls suggest a steady shift in support from PM Sanchez's centre-left party (PSOE) to the alternative centre-right and fiscally conservative People's Party (PP), although they also point to a non-negligible likelihood that the 2027 election will deliver another divided parliament unable to form a clear and stable majority (Exhibit 12).

Exhibit 12: Polls Point to Likely Change of Ruling Coalition in 2027, But Risk of Political Fragmentation Remains



Source: Goldman Sachs Global Investment Research, Politico

Looking ahead to the 2027 election, investors are likely to focus on the likelihood of another parliament without a clear majority and the possibility of a minority government. Prolonged political uncertainty would likely reduce investors' confidence in the continuing transformation of the Spanish economy and weaken the case for our constructive outlook.

Filippo Taddei

Fausto Martin*

* Fausto is an intern with the European Economics Team

The European Economics Team

Sven Jari Stehn
+44(20)7774-8061
jari.stehn@gs.com
Goldman Sachs International

Filippo Taddei
+44(20)7774-5458
filippo.taddei@gs.com
Goldman Sachs International

Alexandre Stott
+33(1)4212-1108
alexandre.stott@gs.com
Goldman Sachs Bank Europe SE - Paris Branch

James Moberly
+44(20)7774-9444
james.r.moberly@gs.com
Goldman Sachs International

Niklas Garnadt
+49(69)7532-1537
niklas.garnadt@gs.com
Goldman Sachs Bank Europe SE

Katya Vashkinskaya
+44(20)7774-4833
katya.vashkinskaya@gs.com
Goldman Sachs International

Giovanni Pierdomenico
+44(20)7051-6807
giovanni.pierdomenico@gs.com
Goldman Sachs International

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Germany: Federal Spending Picked Up Notably in April

Niklas Garnadt
+49(69)7532-1537 | niklas.garnadt@gs.com
Goldman Sachs Bank Europe SE

BOTTOM LINE: German federal expenditures in April came in at EUR 45.8bn, notably above our EUR 42.2bn forecast and EUR 8.0bn higher than in April 2025. Spending was in line with our expectations in the main budget and the military fund and exceeded expectations notably in the infrastructure & climate fund. With revenues falling slightly short of expectations, the fiscal balance was more negative than we expected and remains notably wider than we expected at the start of the year. The fiscal data since the start of the year supports our view of a notable fiscal expansion in 2026, which we expect to provide a fiscal impulse of around 0.6pp.

KEY NUMBERS:

Expenditures in the Main Budget (April): EUR 38.6bn (vs. GS expected EUR 38.3bn and 2025 Realized EUR 36.8bn)

Expenditures (implied) in the Military Fund (April): EUR 1.2bn (vs. GS expected EUR 1.0bn and 2025 Realized EUR 0.9bn)

Expenditures (implied) in the Infrastructure and Climate Fund (April): EUR 6.1bn (vs. GS expected EUR 2.9bn and 2025 Realized EUR 0bn)

MAIN POINTS:

1. German fiscal data for April came in notably above our expectations. Overall spending across the main budget and the off-budget funds for the military and for infrastructure & climate came in at EUR 45.8bn, notably above our expectations of EUR 42.2bn and EUR 8.1bn higher than spending in April 2025.

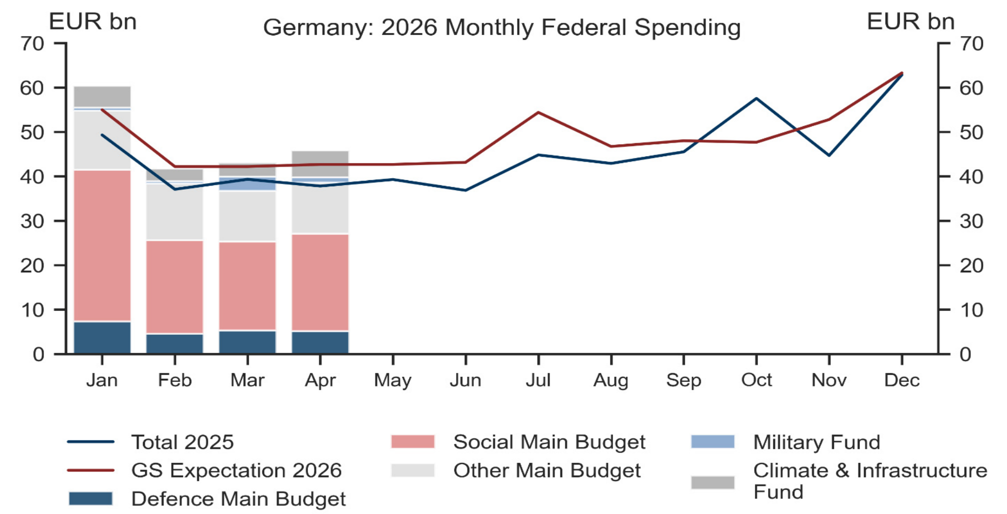
2. Spending stood at EUR 38.6bn in the main budget in April, about EUR 0.3bn above our expectation and EUR 1.8bn above April last year. Both social spending and investment were higher than we expected, while defence spending in the main budget continued to fall slightly short of our expectations.

3. Spending in the off-budget military fund (implied by the fund's issuance activity) stood at EUR 1.2bn, slightly higher than our expectation of EUR 1.0bn and also higher than last year's EUR 0.9bn. YTD spending in the military fund is up EUR 1.7bn (+44%) from last year and total defence spending is up EUR 7.0bn (+34%). Spending in the infrastructure fund (implied by the fund's issuance activity) came in at EUR 6.1bn, notably above our expectation. Data on spending in the federal bucket of this fund, which came in at only EUR 0.7bn in April, suggests that the upside surprise came either from disbursement to the states or spending in the climate transformation fund. As the infrastructure & climate fund was only established last autumn, there are no comparable numbers for last year, but total spending out of this fund remains notably higher than the decrease in infrastructure-related items in the main budget compared to last year.

4. As revenues in the main budget came in below our expectations, the fiscal balance in the main budget was wider than we expected at EUR -9.2bn (vs. EUR -8.0bn expected). Net issuance in the military fund stood at EUR 1.2bn and in the infrastructure & climate fund it stood at EUR 6.1bn, adding up to a total fiscal deficit of EUR -16.4bn in April. The cumulative fiscal deficit through April stood at EUR -63.6bn, EUR 18.8bn wider than we expected at the start of the year and EUR 36.2bn wider than in April last year.

5. The fiscal data since the start of the year supports our view of a notable fiscal expansion in 2026, which we expect to provide a fiscal impulse of around 0.6pp.[1] Due to a larger fiscal deficit year-to-date than we expected, we continue to see some upside risk to our fiscal impulse estimate.

Exhibit 1: Federal Spending in April was Notably Above Our Expectations

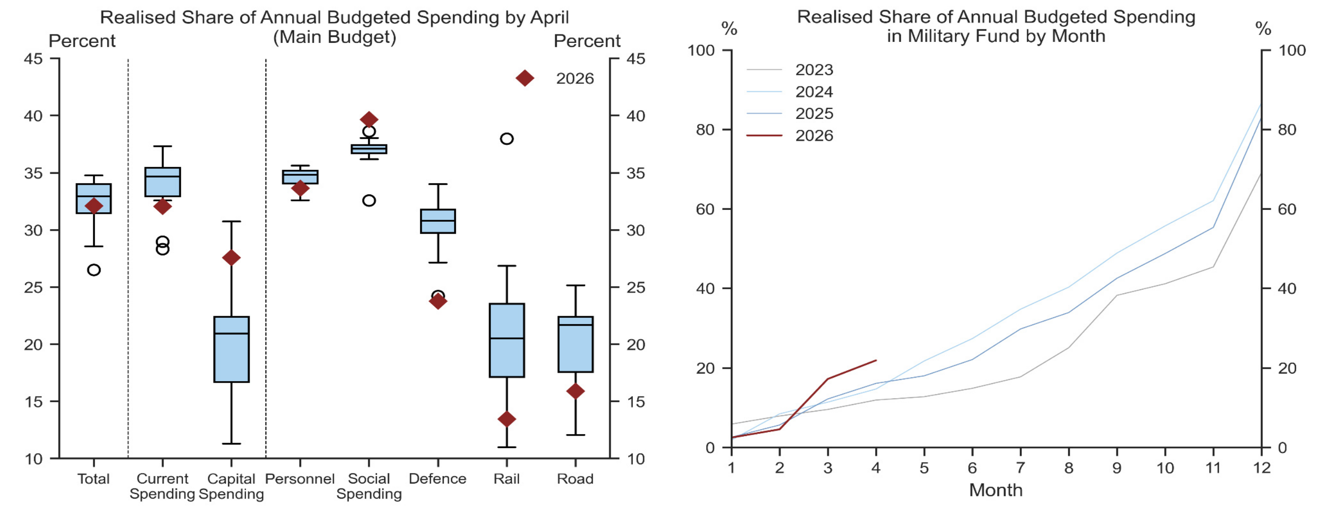


For the fully debt financed military and climate & transformation off-budget funds we infer monthly spending from monthly net issuance.

Source: Haver Analytics, Goldman Sachs Global Investment Research

[1] We recently upgraded our fiscal impulse estimate slightly by 0.1pp to reflect the lower than previously expected 2025 budget deficit.

Exhibit 2: Budget Execution in the Main Budget is in Line with Historical Norms; Spending from the Military Fund has Picked Up Notably



Source: Bundesministerium der Finanzen, Goldman Sachs Global Investment Research

Exhibit 3: Overview of the German Federal Budget for 2026

	2025			2026 FY			Execution Until Apr					Execution in Apr				
	Budgeted (1)	Realized (2)	Share Realized (3)/(2) (3)	Budgeted (4)	GS Expectation (5)	Expected Share (5)/(4) (6)	Realized 2025 (7)	GS Expectation 2026 (8)	Realized 2026 (9)	Realized 2026 vs. 2025 (9)/(7) (10)	Realized vs. GS Expectation (9)/(8) (11)	Realized 2025 (12)	GS Expectation 2026 (13)	Realized 2026 (14)	Realized vs. 2025 (15)	Realized vs. GS Expectation (16)
Main Budget : Revenues	420.7	427.9	102%	426.4	430.0	101%	136.0	136.7	127.3	94%	93%	30.2	30.4	29.4	97%	97%
Main Budget: Spending	502.5	493.3	98%	524.5	510.5	97%	159.5	166.2	168.4	106%	101%	36.8	38.3	38.6	105%	101%
Social Spending	235.9	234.1	99%	245.1	245.0	100%	86.5	90.4	97.2	112%	107%	20.7	20.3	22.0	106%	108%
Defence	69.9	67.0	96%	93.5	87.0	93%	16.9	26.5	22.2	131%	84%	4.2	6.5	5.1	123%	78%
Other	196.7	192.2	98%	185.9	178.5	96%	56.1	49.2	49.0	87%	100%	12.0	11.5	11.5	96%	100%
Memo: Investment	62.7	55.4	88%	58.4	50.9	87%	14.2	13.1	16.1	113%	123%	3.0	2.8	3.4	113%	123%
Main Budget: Fiscal Balance	- 81.9	- 65.4	80%	- 98.1	- 80.5	82%	- 23.5	- 29.5	- 41.0	175%	139%	- 6.6	- 8.0	- 9.2	138%	115%
Off-Budget Funds																
Military	24.1	20.0	83%	25.5	21.5	84%	3.9	3.9	5.6	144%	144%	0.9	1.0	1.2	126%	123%
Climate Transformation	36.7	26.6	72%	34.8	30.1	86%							-			
Climate & Infrastructure	37.2	24.3	65%	58.1	47.6	82%	-	11.4	17.0	-	148%	-	2.9	6.1		
Memo																
Total Spending	590.5	554.2	94%	632.9	599.6	95%										
Total Defence	94.0	87.0	93%	119.0	108.5	91%	20.8	30.4	27.8	134%	91%	5.1	7.5	6.3	123%	84%
Total Investment	115.6	86.8	75%	126.6	104.7	83%										
Total Fiscal Balance	- 147.1	- 109.7	75%	- 183.8	- 149.6	81%	- 27.4	- 44.8	- 63.6	232%	142%	- 7.6	- 11.8	- 16.4	217%	139%

As the climate transformation fund does not publish monthly numbers, total spending, and investment cannot be inferred during the year. We assume the fiscal balance of the climate transformation fund to be 0 throughout the year in line with our expectation.

Source: Bundesministerium der Finanzen, Haver Analytics, Goldman Sachs Global Investment Research

Niklas Garnadt

The European Economics Team

Sven Jari Stehn
+44(20)7774-8061
jari.stehn@gs.com
Goldman Sachs International

Filippo Taddei
+44(20)7774-5458
filippo.taddei@gs.com
Goldman Sachs International

Alexandre Stott
+33(1)4212-1108
alexandre.stott@gs.com
Goldman Sachs Bank Europe SE - Paris Branch

James Moberly
+44(20)7774-9444
james.r.moberly@gs.com
Goldman Sachs International

Niklas Garnadt
+49(69)7532-1537
niklas.garnadt@gs.com
Goldman Sachs Bank Europe SE

Katya Vashkinskaya
+44(20)7774-4833
katya.vashkinskaya@gs.com
Goldman Sachs International

Giovanni Pierdomenico
+44(20)7051-6807
giovanni.pierdomenico@gs.com
Goldman Sachs International

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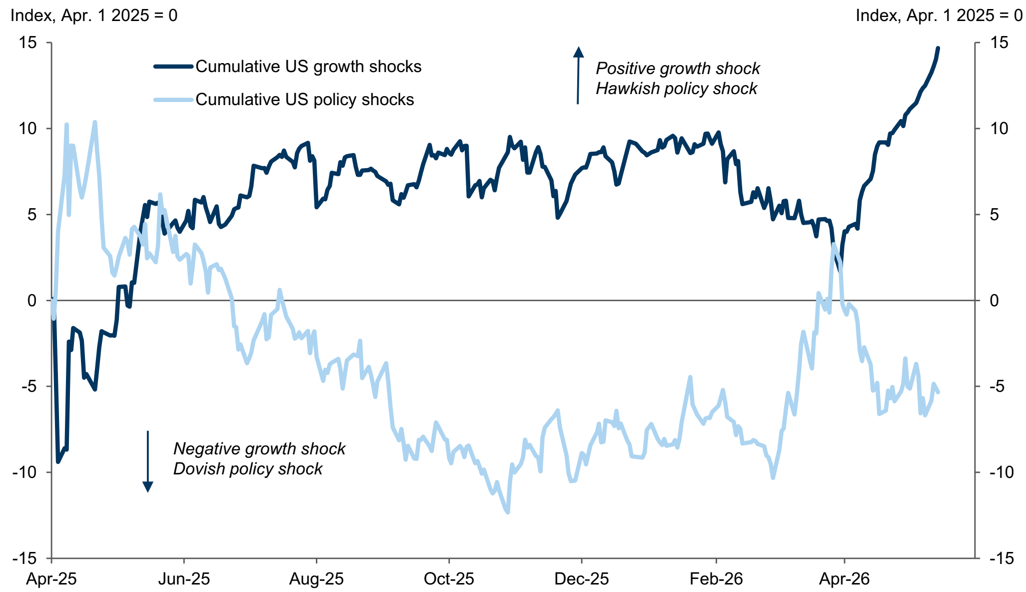
Global Market Views: Shortages

Dominic Wilson
+1(212)902-5924 |
dominic.wilson@gs.com
Goldman Sachs & Co. LLC

Kamakshya Trivedi
+44(20)7051-4005 |
kamakshya.trivedi@gs.com
Goldman Sachs International

1. The risks of relief. The Iran ceasefire and the reduction in deep downside tails have allowed the market to look ahead and compress risk premia across a range of assets. That type of narrowing in the distribution is often associated with sharp market gains in situations of fundamental uncertainty, and this has proved to be the case again. Despite oil prices and yields staying elevated, a broad range of assets—including US equities, high carry/commodity FX, and EM—have recovered smartly, and renewed AI optimism has pushed those exposures to new cycle highs. The common element is that shortages—in commodities and the AI supply chain—are driving capital flows. After a sharp recovery, the distribution of risks is more balanced. The most obvious downside threat is still a re-escalation of hostilities in the Middle East or a prolonged Hormuz closure that raises the probability of even higher oil prices, higher rates, lower growth and recession. As we wait, the flip side of shortages alongside more resilient growth is upward pressure on bond yields, which the market tends to worry about more when they break new ground. By contrast, even a gradual restart of energy flows should allow some relief across oil prices and rates markets that are now quite hawkishly priced, and lead to a broadening of the positive price action geographically and across sectors. Beyond that, we are in many ways back to the tensions we highlighted in our outlook—between well-valued assets and increasing concentration on the one hand (with AI-related exposures a poster-child for that), and a macro cycle that does not yet show the imbalances and leverage excess that is typical of late-cycle behaviour. Those tensions mean that even if we avoid the worst tails, we are likely to see volatility rise alongside further equity price increases.

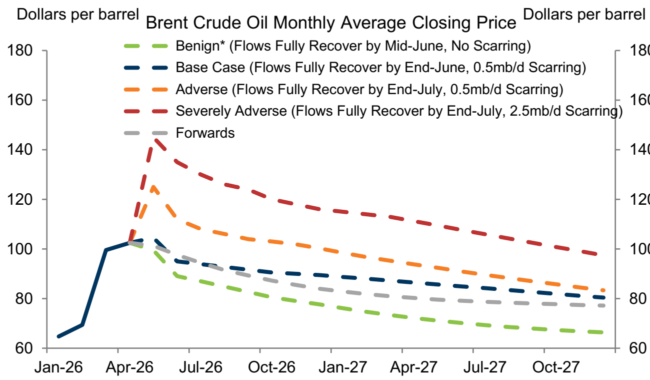
Exhibit 1: Sharp relief in growth worries, policy shock lingers



Source: Goldman Sachs Global Investment Research

2. Iran re-escalation still the main risk. The biggest risk to the macro picture still comes from a failure to resolve the Iran war and the flow of oil. Equities and other risk assets have priced in substantial relief on the basis that the deep tail risks are avoided. While that assumption holds, markets have been more tolerant of headline risk. But the flip side to this higher threshold for worry is that a more serious challenge to the market's optimism could lead to a larger repricing. Although we agree that the risks have fallen, the prospect of more adverse outcomes is still very real, and we think that deeper downside tail is underpriced. There is a challenging circularity here too: although markets are looking through any temporary disruptions, it may also be that another bout of market worry is needed to force an agreement that allows oil flows to resume. The longer we go without a clear peace agreement and a convincing reopening of the Strait of Hormuz, the more likely we are to revisit that risk as energy product shortages become clearer. We continue to think there is value in hedges against this outcome. Out-of-the money downside in European equities, credit and FX still screens best in our cross-asset comparisons. Longs in oil may also continue to be protective, though are also vulnerable to a full resolution of the crisis.

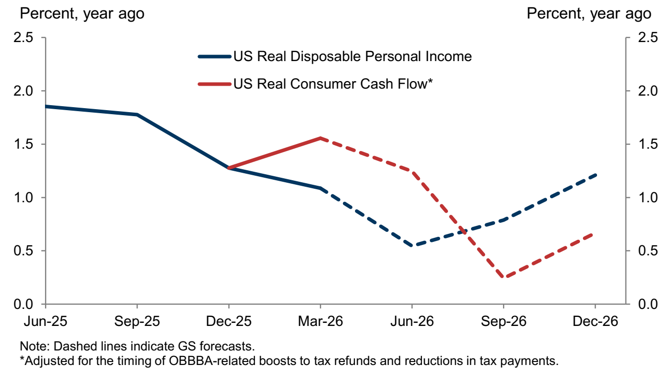
Exhibit 2: Oil price risks still skewed to the upside



*Benign scenario includes higher US and core OPEC production.

Source: Goldman Sachs Global Investment Research

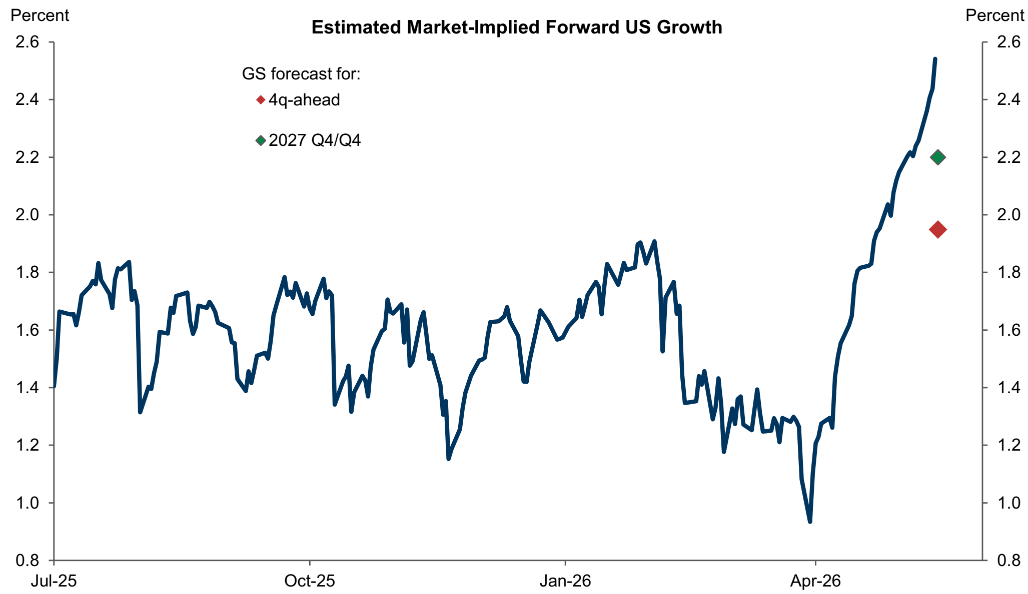
Exhibit 3: Real income softness ahead for the US



Source: Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

3. Macro optimism already well priced. The market has already priced significant relief. Our standard estimates of US growth pricing from the joint shifts in equities and bonds now stand at 2.5%. These measures may overstate the true cyclical optimism, since tech and commodity strength may be boosting them for other reasons, but they are consistent with a market that is already looking through the short-term weakness we expect to the better growth outlook in 2027, and is now potentially overshooting those levels. On our forecasts, there is room for some policy relief in the months ahead (our rate forecasts are generally dovish to market pricing) conditional on an Iran resolution. But with the US economy and labor market holding up better than widely expected, and inflation likely to be elevated at least in the short term, the scope for near-term policy relaxation is likely limited unless there is a very clear improvement in oil flows and an end to the war. Part of the reason why our measures of growth pricing have risen sharply is that bond yields and equities have been rising steadily together. Markets are now beginning to question whether that combination is sustainable. We think that inflation pressure is likely to begin to fade within a couple of months of the energy price peak and the market is already priced for hikes, including now in the US. So, while we may well see further near-term pressure for higher yields, we think that pressure will ultimately be contained. But given how relaxed equity markets have been about rising rates so far, markets may continue to worry in the coming weeks that a mix of more hawkish policy pricing and less bullish growth pricing may be needed.

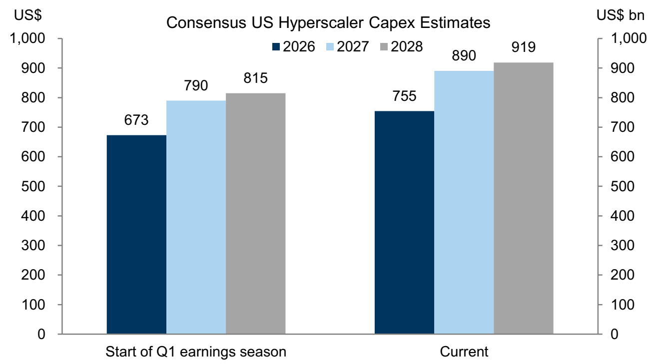
Exhibit 4: Market pricing of US growth already moving past our 2027 growth forecasts



Source: Goldman Sachs Global Investment Research

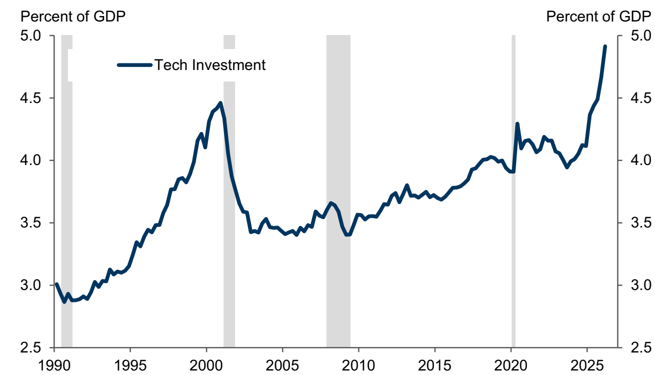
4. Back to AI amid booming capex. The AI theme is back in force and has again been a focal point for markets. Outside energy-related areas, the markets that have moved most decisively through their pre-war highs are AI-heavy indices such as Korea, Taiwan and Nasdaq. As in 2025, concerns in Q1 about the sustainability of the AI boom have dissipated in Q2 in a way that has overlapped with, but is separate to, a broader macro shock. The value creation here continues to favor heavily those involved in supplying the AI investment boom, especially in the semis/memory spaces where shortages are clear. Capex projections have moved sharply higher, driving positive earnings revisions across those beneficiaries, and tech investment spending has now exceeded the peaks from the late 1990s as a share of GDP. But with profits holding up (corporate profits as a share of GDP are also at their highest on record), most of the macro imbalances that signaled growing trouble in the late 1990s are still not visible yet. Worries about private credit, which we think are unlikely to have systemic consequences, have also settled a little.

Exhibit 5: Another sharp rise in hyperscaler capex expectations



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 6: Tech investment already at record levels

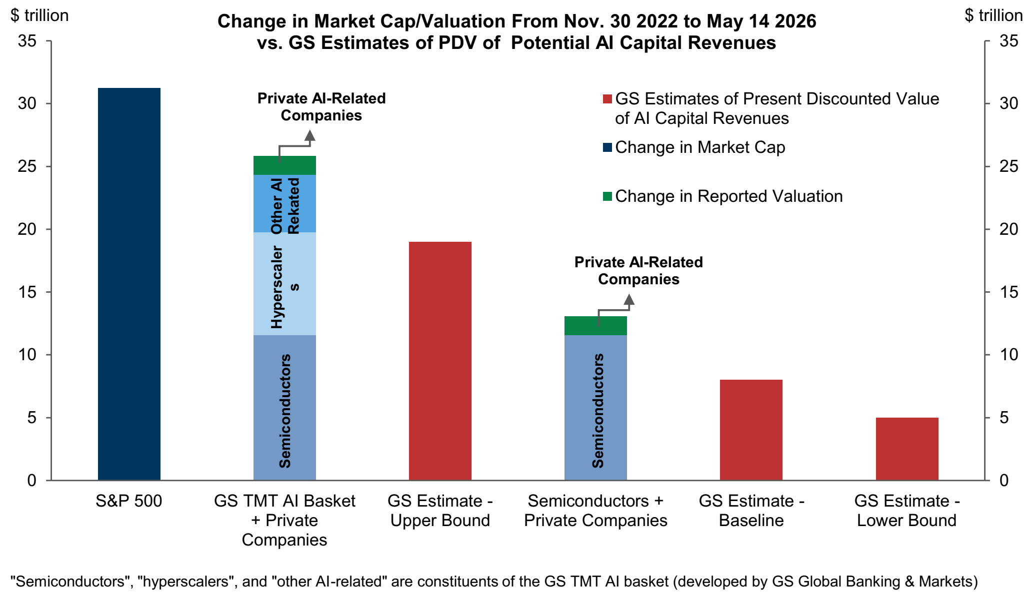


Shading indicates NBER recessions. Tech investment includes private nonresidential fixed investment in software and information processing equipment.

Source: Haver Analytics, Goldman Sachs Global Investment Research

5. The value challenge. As we highlighted in our outlook for 2026, the AI theme is a poster child for the broader tension between a macro cycle that (ex the Iran shock) still looks like it has room to run and market pricing that already looks quite generous. The latest rally has continued to drive the cumulative value embedded in AI-related companies higher. And while not all of this is directly attributable to AI, the market is taking credit for substantial macro benefits. This increases the vulnerability to worries about whether the benefits will be justified and whether they are appropriately allocated (in particular, whether it is sustainable for them to accrue so heavily to the chip layer). There has been more focus on potential winners and losers this year—and dispersion has risen. But the risk is that the market falls into either a fallacy of aggregation (assuming that more individual companies can be winners than the economy overall can deliver) or of extrapolation (assuming that earnings supported by the investment boom itself can be sustained). As long as earnings and spending plans continue to rise more quickly than expected, there is likely to be a tailwind to the AI complex. But in the process, the market may build a valuation overhang that may need to be resolved at some point.

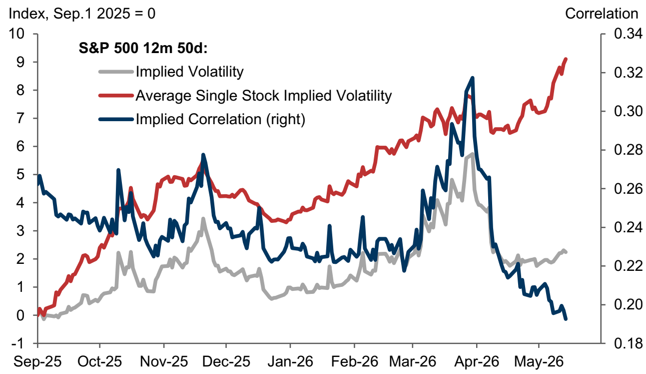
Exhibit 7: Even more value built into AI-related areas



Source: Bloomberg, FactSet, Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

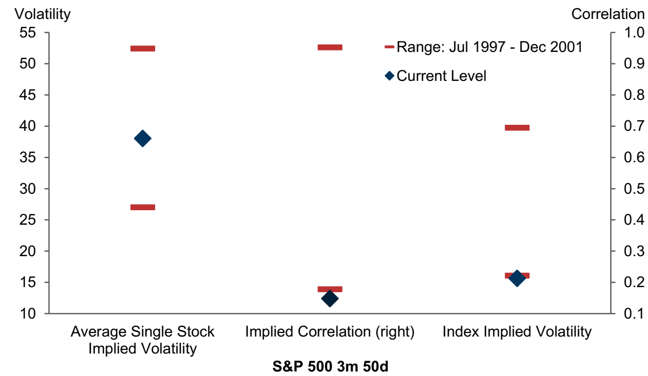
6. Higher volatility, but not yet broad-based. One of our core views for 2026 was that longer-dated equity volatility was likely to rise structurally, as volatility around the AI theme and its financing or late-cycle macro dynamics became more prominent. Since last September, there has been a clear upward drift in longer-dated implied volatility in the SPX, despite a sharp rally in equities over that period. But the rise in average single stock volatility—and in concentrated indices like Korea—has been much larger and more persistent than the index story. We think that this is partly because there has been more focus on "distributional" volatility within the AI theme (winners versus losers) than on "macro" volatility (the aggregate value). This has raised single stock volatility but pushed correlations to record lows, limiting the upward pressure on volatility in broad-based indices. We still think that there are good reasons for index volatility to rise over time and any macro shock should drive correlations higher. So we still like pairing long equity views with longs in longer-dated SPX volatility and see longer-dated equity options as appealing ways to limit losses while retaining exposure to further equity upside. And we generally like the idea of owning volatility across assets, whenever it cheapens meaningfully. But the mix of high single stock volatility and low correlation—which we saw at points in the late 1990s too—does mean that index volatility may rise less than expected in some of these shocks.

Exhibit 8: Rise in single stock volatility more persistent than at the index level



Source: Goldman Sachs FICC and Equities, Goldman Sachs

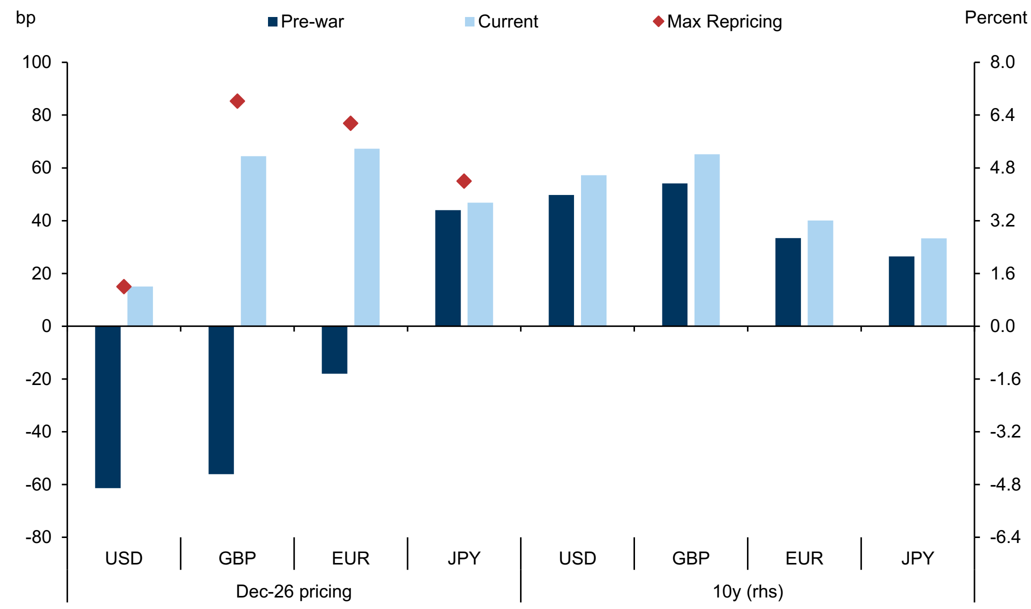
Exhibit 9: Index volatility and correlations still low versus the late 1990s



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

7. Rates running out of time. While many assets have retraced fully and beyond, the hawkish shifts across rates market pricing have generally held. As we move through this messy stalemate where the Strait of Hormuz is still closed, energy prices are still high, growth is holding up, but the first signs of higher prices are coming through in inflation prints, it is possible that the market re-tests or surpasses some of the highs in front-end rate pricing. We continue to think that the threshold for hikes in the US is still very high, or, for that matter, extended hikes elsewhere. But the bulge in inflation without labor market weakening clearly makes easing harder too. Certainly relative to our dovish views coming into the year, we now expect fewer or no cuts across the world, with markets having moved even more hawkishly relative to that. This leaves the forward outlook more balanced than before the war, with more risk of hikes reflected in market pricing. If the energy supply situation resolves, and if the sequential inflation bulge moves through quite quickly, there is again scope for rate relief, particularly front-ends. So, while our bias continues to be that the hikes priced across many DM and EM front-ends are unlikely to be matched, we are also running out of time for cuts in 2026. Linked to that, terminal and longer-dated rates have also been dragged higher by the Iran crisis, and by higher term premia in the UK and Japan. Those moves are also at odds with prior worries about AI displacement and labor market slack, so there too there may be scope for a modest reprieve in the event of an Iran resolution; but it is likely to be limited given ongoing fiscal expenditures (on defense, energy security, or social support) and private spending (on the AI infrastructure build-out).

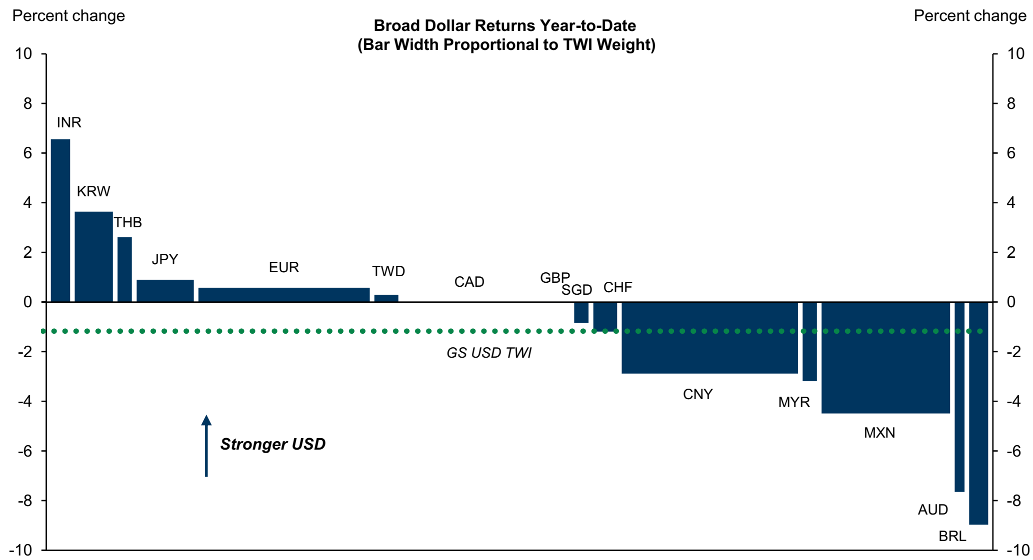
Exhibit 10: Hawkish repricing has extended from front-ends to long-end



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

8. Dollar stability masks deeper FX shifts. The reversal in the Dollar since the ceasefire means that it is now largely unchanged, both since the Iran crisis, and year-to-date. Shifts in the Dollar—both up and back down— have been mostly in line with the moves implied by the beta to commodities and risk assets. In that limited sense, Dollar correlations have behaved as expected through this shock so far, and we do not see any new impulse to De-Dollarisation in its broad stability. That stability does, however, mask other significant trends playing out across global FX. Near-neighbour crosses like NOK/SEK (Norwegian versus Swedish Krona) and AUD/NZD (Australian versus New Zealand Dollar) have moved much further to reflect commodity terms-of-trade differences and are now at multi-year extremes. EM FX carry has also continued to generate strong total returns, both versus the Dollar but especially against lower-yielding funders such as the Euro, Swedish Krona and Thai Baht. Potentially even more important is the trend towards sustained CNY appreciation, that has continued through this period. While the Trump-Xi summit could prove a catalyst for more CNY strength, we think that the case for a stronger Renminbi is more fundamental and longer-lasting. China's external surplus is approaching unprecedented levels as a share of global GDP, reflecting deep levels of export competitiveness, and also a highly undervalued currency, with currency appreciation an equilibrium outcome. The Japanese Yen is also very cheap, and while fresh bouts of intervention are a reminder of that reality, we don't see sustained strengthening unless there are stronger concerns about a recession (with lower equities and rates) or there is a more meaningfully hawkish pivot from the BoJ.

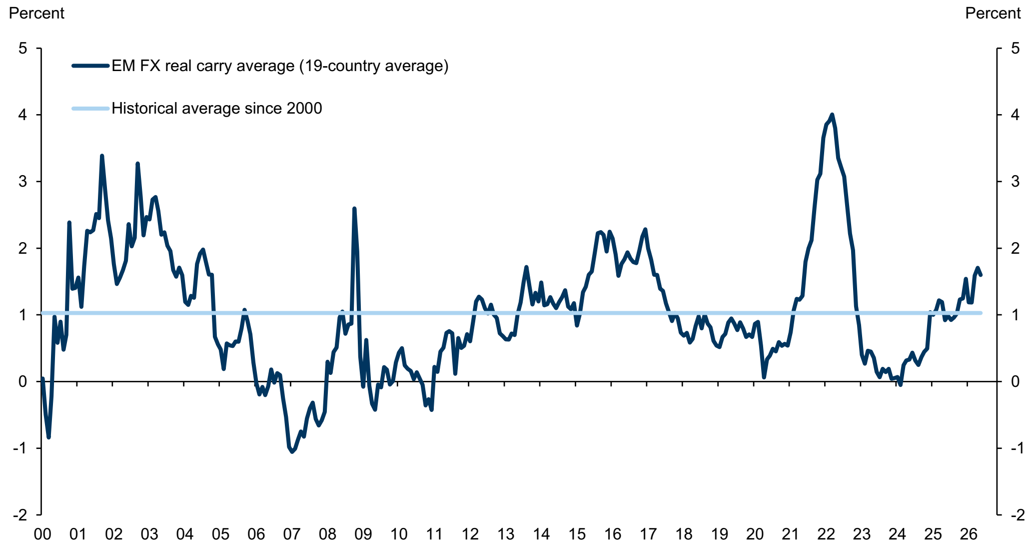
Exhibit 11: Broad Dollar stability masks large moves



Source: Bloomberg, Goldman Sachs Global Investment Research

9. EM: Carry on waiting. The EM equity complex, with its exposure to the memory cycle via Korea and Taiwan, and to commodities via LatAm, has managed to generate strong outperformance YTD despite the energy price volatility. That barbell should remain well-suited to the current environment even as the North Asia (long 'chips') versus South Asia (short 'oil consumers') differentiation becomes even more stretched. But EM carry has been an additional outperformer as we wait in a zone between no clear deal to re-open the Strait of Hormuz but also no kinetic hostilities. Much of this outperformance relates to the fact that many high-carry EM currencies—such as BRL, MXN and COP—also happen to be energy producers. But there has been another underappreciated factor that should lengthen the window in which EM carry can perform. Even if some of the rate hikes priced in DM and EM curves are not delivered, the bar is now higher for rate cuts as Central Banks 'wait and see' whether energy supplies are restored, price shocks are indeed transitory, and second-round effects are avoided. As a result, not only is the level of EM carry now higher than it was at the start of the year and high relative to long-term averages, we also now expect a slower decline in real rates and carry levels relative to the start of the year. Taking into account carry levels, relative terms-of-trade shifts and idiosyncratic developments, we prefer BRL, HUF, MXN and ZAR as Longs for an EM Carry Basket. More diversified funders (including EUR, SEK and THB, apart from the Dollar) offer higher carry returns alongside added resilience to energy price and risk sentiment shifts.

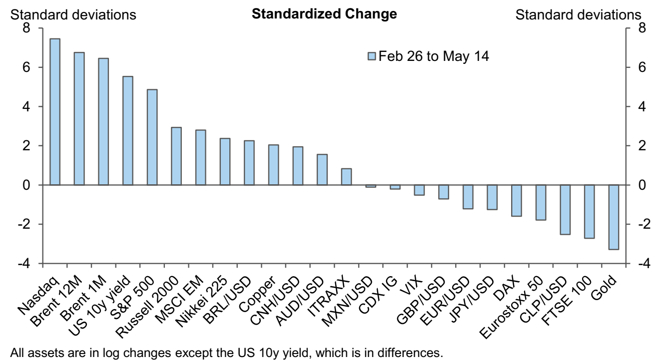
Exhibit 12: EM carry has edged higher YTD and is higher than long-term averages



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

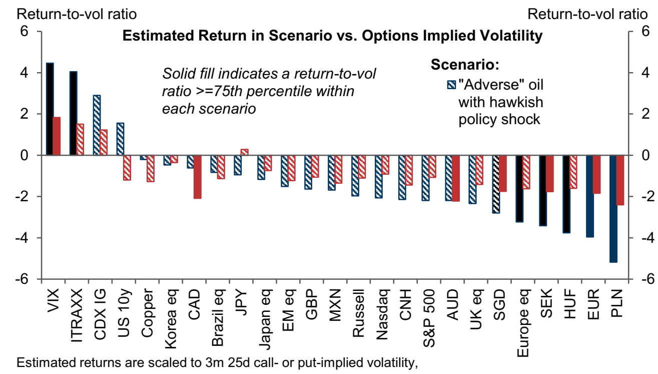
10. Shortages and beyond. Even as the escalatory tail remains crimped, a prolonged messy ceasefire still creates opportunities and risks to navigate. A simple framework for some of the sharpest asset price moves we see is that they reflect physical shortages: prices need to rise sharply to signal that more capital is needed and to incentivize it to flow there. That includes memory chips, power and—at least for now—energy products. Energy resilience is also likely to be a growing theme driving capital spending into renewables across oil-deficient Asia and Europe. Until some of those shortages are resolved, say because new, cheaper sources of supply become available, those moves may be sustained. The counterpart to this is that booming AI investment, fiscal spending, including on defense, and resilient commodity markets may keep bond yields higher than normal. Away from these areas, EM carry strategies can continue to perform against a range-bound Dollar backdrop, and we think it makes sense to retain exposure to the growing divergence between energy exporters/producers and importers/consumers in both equities and currencies as long as full resolution is some way off. In the event of a fuller resolution of the Iran war, the path for risk assets should be further buffered by rate relief (as hike pricing relaxes) and the fact that downside growth tails (from the energy constraint) are set aside. But it is also likely to be punctuated by bouts of volatility given optimistic growth pricing, concentrated positioning, and risks to long-end rates (especially if growth damage is less than feared). So we continue to advocate for meaningful hedges of the downside tails, especially in the shorter term. Renewed worries about credit and equity markets digesting AI-related issuance, returns on those investments, and the aggregation/extrapolation fallacies also argue for pairing constructive risk positions with long volatility exposure to protect against those reassessments and market reversals.

Exhibit 13: Best-performing assets mostly linked to either AI or energy shortages



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Exhibit 14: We continue to like hedging for downside tails, with downside in European assets still screening well



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

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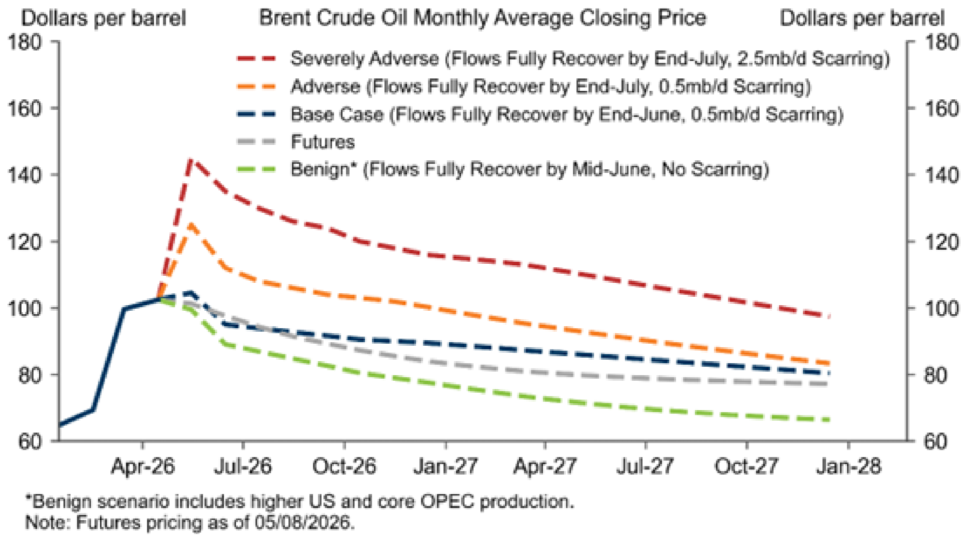
Global Views: Bending, Not Breaking

Jan Hatzius
+1(212)902-0394 | jan.hatzius@gs.com
Goldman Sachs & Co. LLC

1. We see three reasons why the 10-week closure of the Strait of Hormuz has hurt global growth only moderately so far. First, oil prices have not risen as much as expected, partly because of unusually high pre-war inventories and partly because markets never lost faith that very large consumer price hikes would prompt a US policy shift. Second, physical shortages in areas such as jet fuel have so far been met with relatively painless forms of demand destruction (e.g. a large shift to renewables in China and reduced flight schedules on lower-value routes globally). Third, fiscal policy, the AI boom, and—with a brief interruption in March—financial conditions have been supportive all year. Under our baseline assumption of a gradual Strait reopening that starts soon and finishes in late June, we see Brent prices stable in the near term and edging down to $90/barrel by year-end. However, the risks remain tilted toward more adverse outcomes, higher oil prices, and greater economic damage.

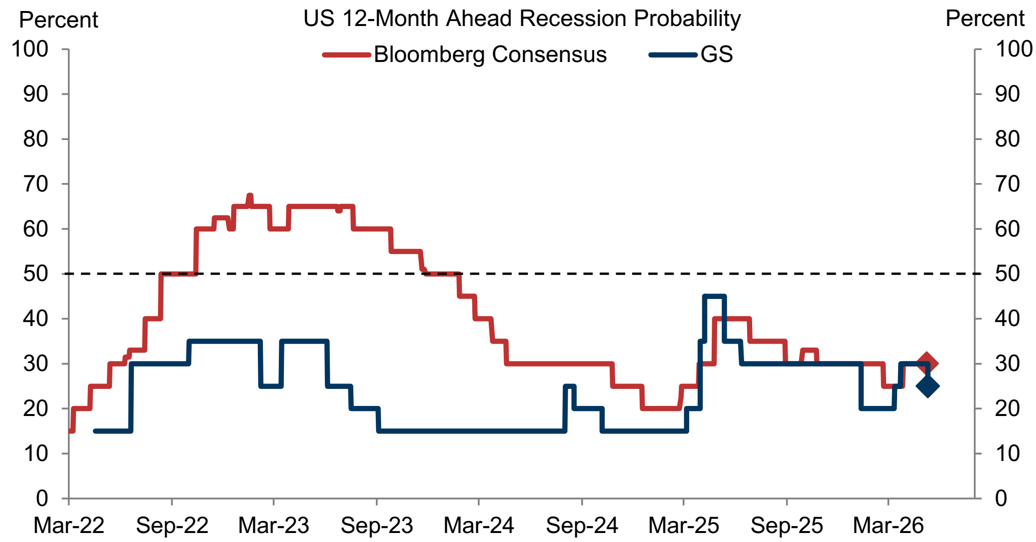
Exhibit 1: Oil Price Risks Still Skewed to the Upside



Source: Goldman Sachs Global Investment Research

2. We have shaved our 12-month US recession probability from 30% to 25% as economic activity has held up well and our financial conditions index has eased back below pre-war levels. While headline GDP growth fell short of expectations in Q1, private domestic final sales grew at a solid 2.5% in real terms. Moreover, the labor market news has mostly beaten expectations, with a bigger-than-expected 115k increase in nonfarm payrolls in April, a stable 4.3% unemployment rate, and a sharp decline in jobless claims.

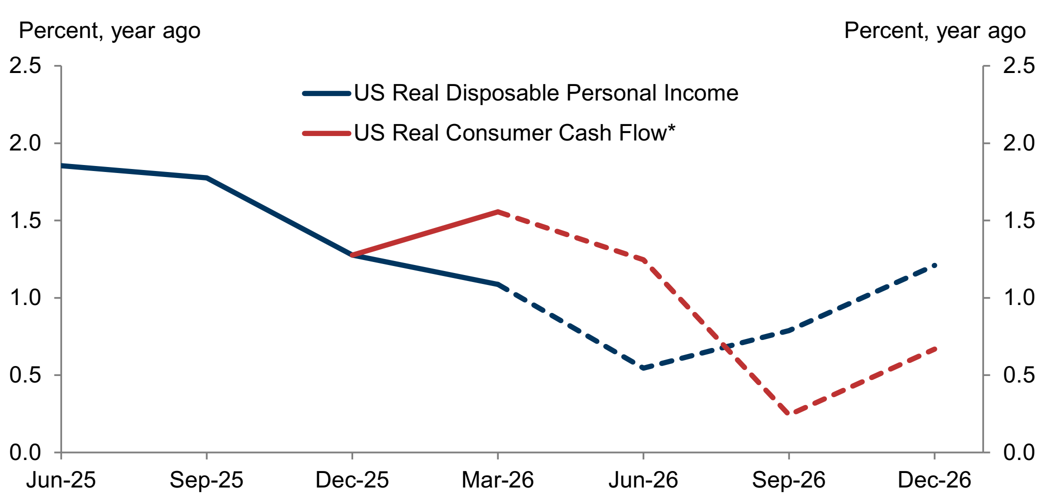
Exhibit 2: Recession Risk Looks Incrementally Lower



Source: Bloomberg, Goldman Sachs Global Investment Research

3. But our recession risk estimate remains 5pp above the pre-war level and 10pp above the unconditional long-term average. Moreover, our baseline forecast entails below-trend growth in coming quarters, mainly because consumers face weakness in real disposable personal income and real cash flow. The boost to cash flow from strong tax refunds is about to fade, and the drags from higher gas prices, slowing nominal wage gains, and reduced healthcare subsidies are likely to persist. With the personal saving rate already down to 3.6%, the lowest level in three years, weaker income and cash flow is likely to translate into weaker consumer spending.

Exhibit 3: Real Income and Cash Flow Are Likely to Soften in H2



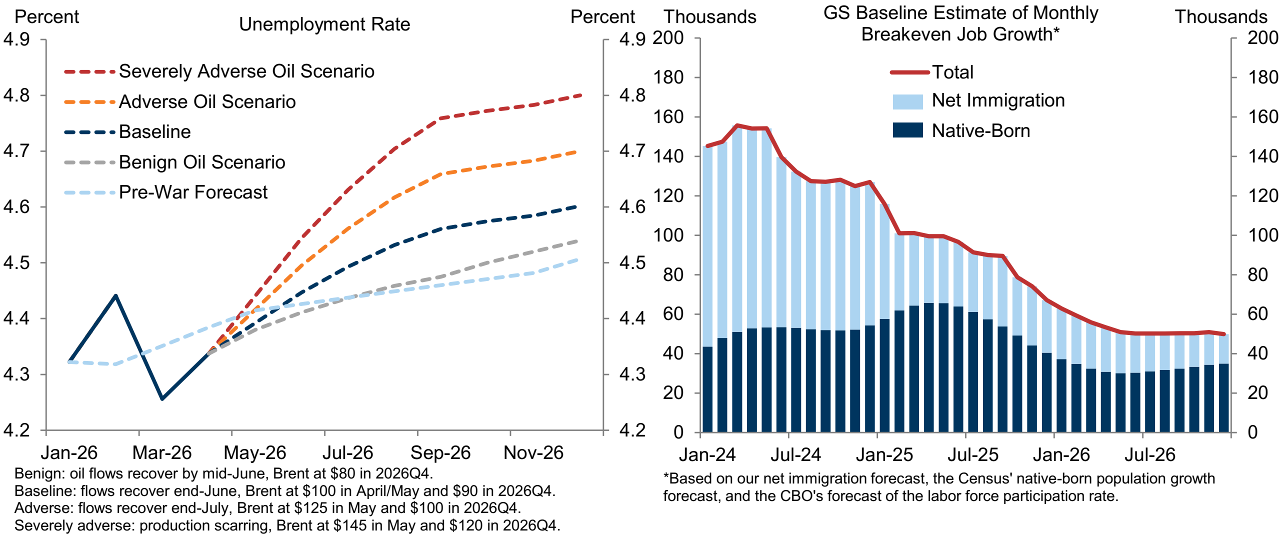
Note: Dashed lines indicate GS forecasts.
*Adjusted for the timing of OBBBA-related boosts to tax refunds and reductions in tax payments.

Source: Haver Analytics, Bureau of Economic Analysis, Goldman Sachs Global Investment Research

4. Beyond the weaker growth outlook, there are two other reasons why the US unemployment rate is likely to drift higher. First, US productivity has already accelerated from a 1.5% trend before the 2020 pandemic to 2.1% since then, and the improvement

is likely to continue as the AI impact broadens. This means fewer new jobs for any given GDP increase. Second, we believe the Fed staff and many private-sector analysts have overshot in revising down their estimates of the "breakeven" rate of payroll growth to around zero. Our estimate of around 50k is consistent with the fact that the unemployment rate has fallen by only a few basis points on net this year, despite average payroll growth currently reported at 76k.

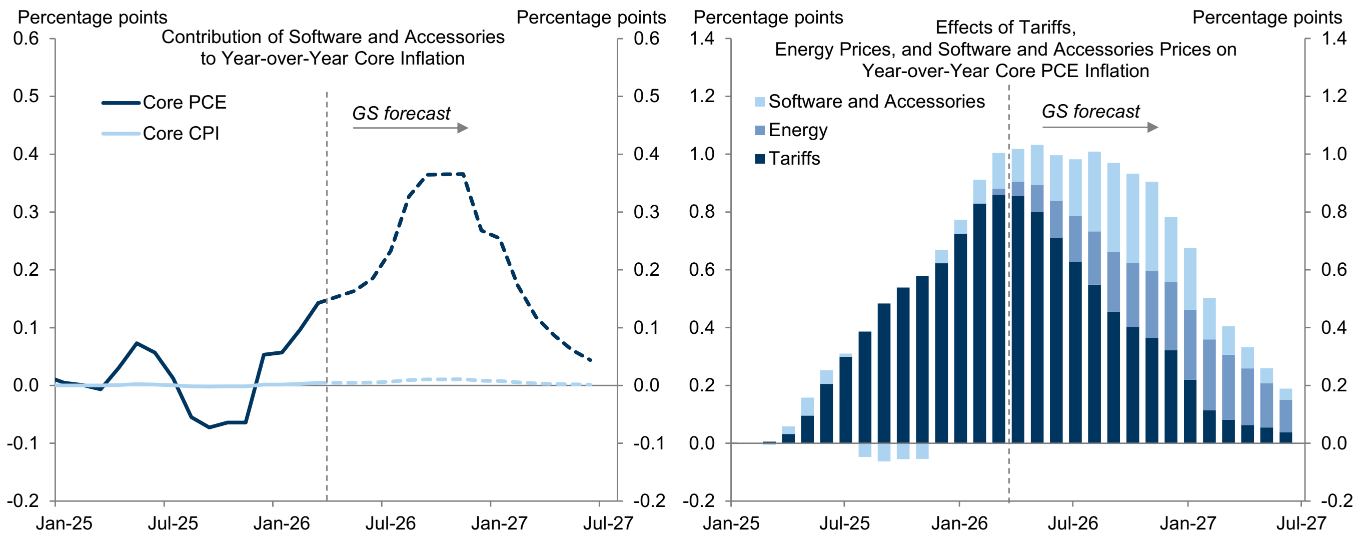
Exhibit 4: Unemployment Likely to Rise, in Part Because Breakeven Payroll Pace Remains Positive



Source: Goldman Sachs Global Investment Research

5. Two distortions related to the AI boom have worsened the near-term outlook for core PCE inflation. First, after adding new AI features to software packages such as MS Office, Microsoft and other firms now charge a higher price for the new bundle than for the prior standalone package. The official statistics don't adjust for the expanded offering and thus incorrectly treat this as a pure price increase. Second, the combined weight of software and accessories—where prices have also surged on the back of costlier memory—is 30x higher in PCE than CPI, apparently because the underlying survey incorrectly captures not only consumer demand but also some demand from small businesses. Together, these distortions are likely to boost the year-on-year rate of core PCE inflation by 0.4pp later this year. Combined with an estimated 0.3pp pass-through from higher energy prices, this will delay the renewed core PCE disinflation we have been expecting.

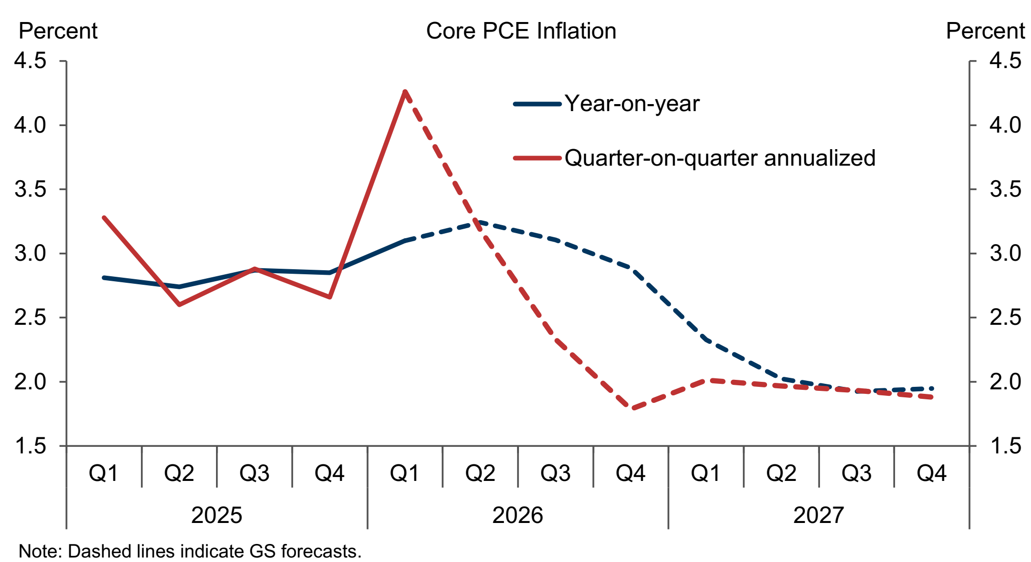
Exhibit 5: Software Is Yet Another Transitory Inflation Shock



Source: Haver Analytics, Goldman Sachs Global Investment Research

6. The longer-term outlook for core PCE inflation is better. While the boosts from tariffs, energy, and software/accessories should fade over the next year, two key drags are likely to remain in place. First, rent inflation should continue to slow as the official CPI/PCE shelter data catch up with market rents, which remain weak because investors overbuilt apartments in the immediate post-pandemic era in several markets (especially in the Sun Belt). Second, the labor market is a source of disinflation, with nominal wage growth of 3.6% year-on-year based on our GS wage tracker and unit labor costs growing just 1.2% year-on-year in Q1. We therefore see sequential core PCE inflation returning to 2% later in 2026, though the year-on-year rate is unlikely to get back to target until mid-2027.

Exhibit 6: Core PCE Inflation Should Look Better by Year-end, Especially in Sequential Terms

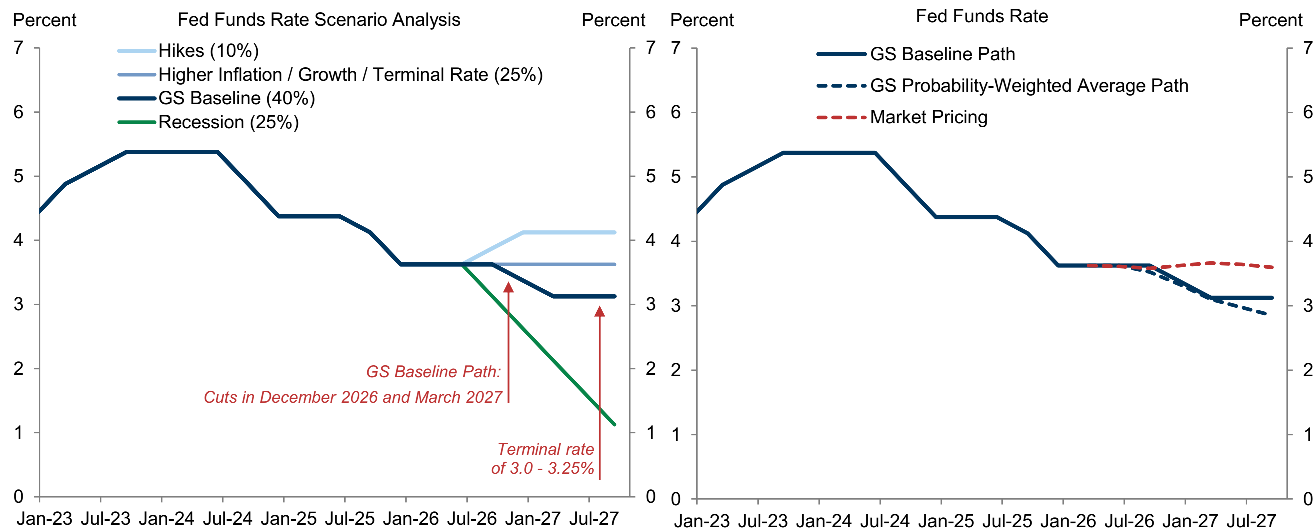


Source: Haver Analytics, Goldman Sachs Global Investment Research

7. Upside PCE inflation surprises, better labor market data, and growing hawkishness within the FOMC have led us to delay our forecast for two more 25bp normalization cuts

to December 2026 and March 2027. However, our forecast remains below market pricing, especially in probability-weighted terms. One important question will be the Fed leadership transition. We doubt the committee would endorse Kevin Warsh's earlier call for cuts in anticipation of AI-driven disinflation. But his more recent argument that trimmed-mean PCE—which is running 0.8pp below core PCE—is a better measure because it is less sensitive to one-time price shocks seems more promising. After five years of above-target inflation, most sitting Fed officials are reluctant to make this argument publicly, lest they be accused of cherry-picking. But many probably agree with it tacitly and might be persuaded to do so more openly alongside a new Chair.

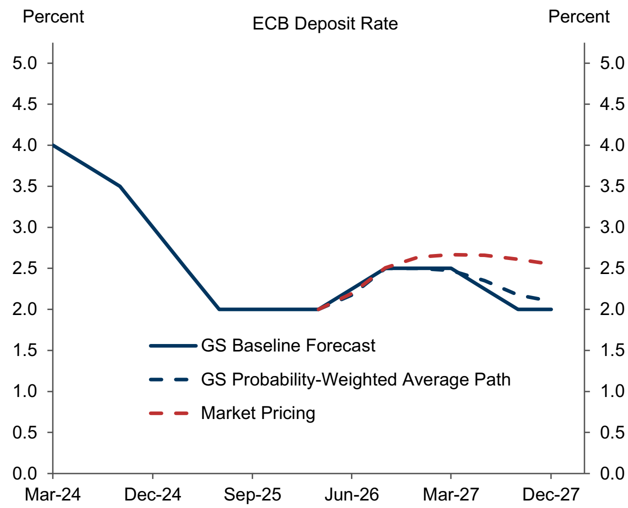
Exhibit 7: Fed Cuts on a More Delayed Schedule

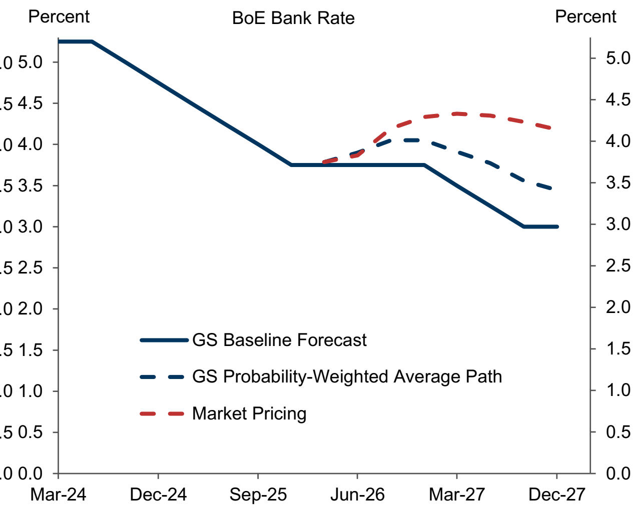


Source: Bloomberg, Goldman Sachs Global Investment Research

8. We still expect the ECB to deliver two 25bp hikes soon—probably in June and September—because of the low 2% starting point for the deposit rate and the historical tendency to (over)react hawkishly to oil-driven headline inflation shocks. However, the risk is on the side of no hikes because core inflation remains very well-behaved so far and ECB officials as a group are more open-minded about core vs. headline than in the pre-Draghi days. In any case, we think that any near-term hikes would probably be unwound in 2027. Across the Channel, we don't see hikes from the BoE in our baseline—in part because Bank Rate at 3.75% is clearly in restrictive territory—but they do constitute a meaningful hawkish risk.

Exhibit 8: We See the ECB Hiking and the BoE on Hold, But Our Views Are Dovish Relative to Market Pricing in Both Cases

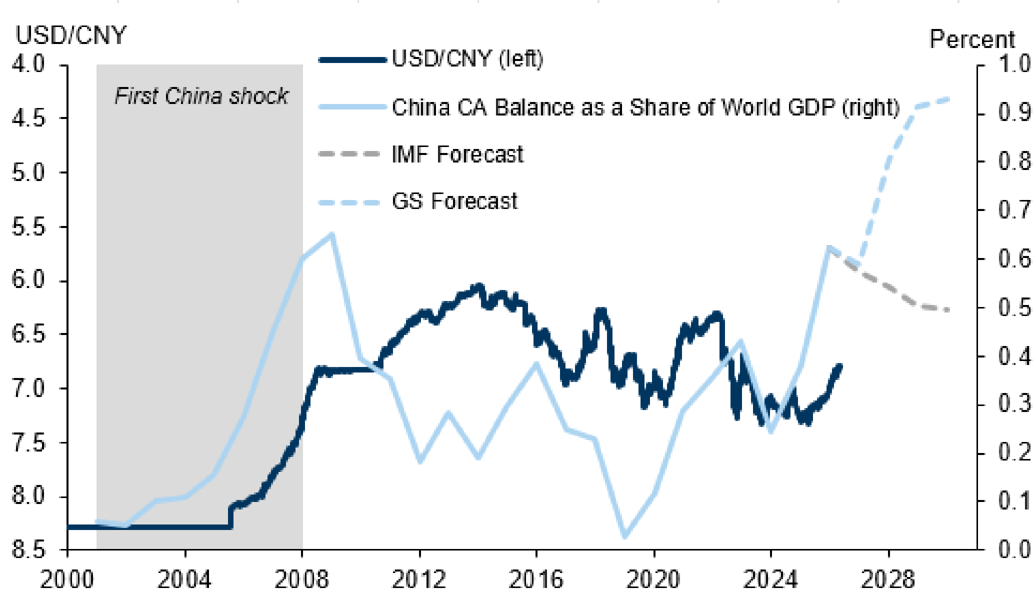




Source: Bloomberg, Goldman Sachs Global Investment Research

9. In the run-up to this week's Trump-Xi summit, China's economy continues to benefit from strength in the export-oriented goods sector. This kept Q1 GDP growth at 5.0% year-on-year and our 2026 forecast at 4.7%, but the economy remains very unbalanced. While higher prices for imported energy should temporarily halt the increase in the current account surplus this year, we expect the combination of strong production and sluggish demand to push it further into unprecedented territory as a share of global GDP in subsequent years. The clearest market implication is that CNY is undervalued, especially against the US dollar where our FX strategists estimate a gap of more than 20% and have doubled down on their forecast of significant appreciation, with a new target of 6.50 in 12 months.

Exhibit 9: China's Record Current Account Surplus Should Boost the CNY



Source: Goldman Sachs FICC and Equities, Haver Analytics, IMF, Goldman Sachs Global Investment Research

10. If we wrote down the most frequent topics in discussions with market participants, the vast majority would be negative—an unprecedented oil supply shock, large fiscal

deficits, questions about Fed independence, population shrinkage, cyber/AI security risks, US-China rivalry, and the biggest war in Europe since WW2, amongst others. Moreover, equities are far from cheap. So why have they performed so well? The likely answer is the strength of the Q1 earnings season combined with the improving long-term outlook for productivity and thus profits on the back of the AI revolution. The latter is especially important in the US market, where our strategists estimate that earnings and dividends a decade or more in the future represent roughly 75% of fundamental value. That said, a market where the baseline is positive but the risks are so asymmetrically negative argues for combining long risk position with an exposure to equity volatility or significant hedges (our cross-asset strategists recommend equity puts struck deeply out of the money).

Jan Hatzius

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Reg AC

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